Image Processing Fundamentals

Dan Wu
danwu.bme@zju.edu.cn

Previously…

- Digital Images and Biomedical Images
- Format of Digital Images
- What is Image Processing?
- Human Vision System

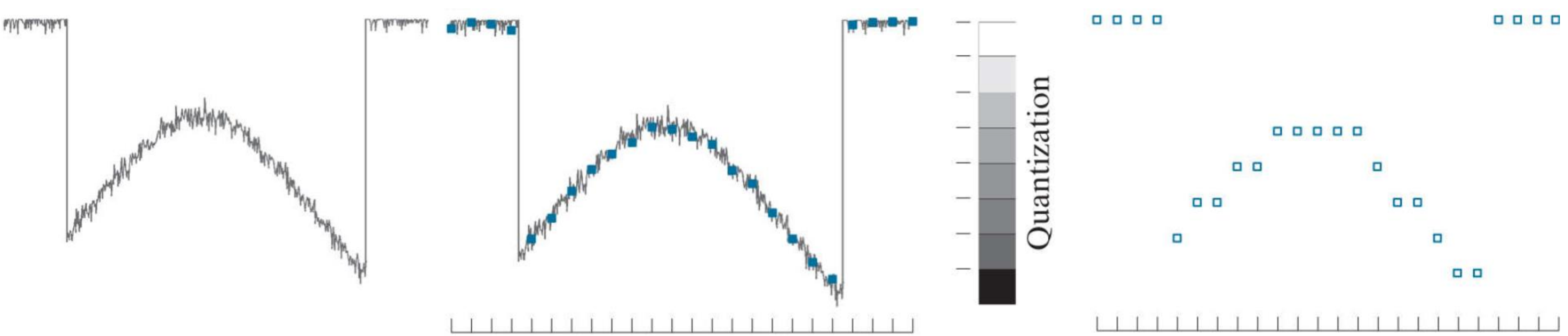
Outline for today

- Basic BIP Concepts (Chapter 2.4-2.6)
  - Sampling and quantization
  - Interpolation

- Color Image Processing (Chapter 7)
  - Color model
  - Pseudocolor

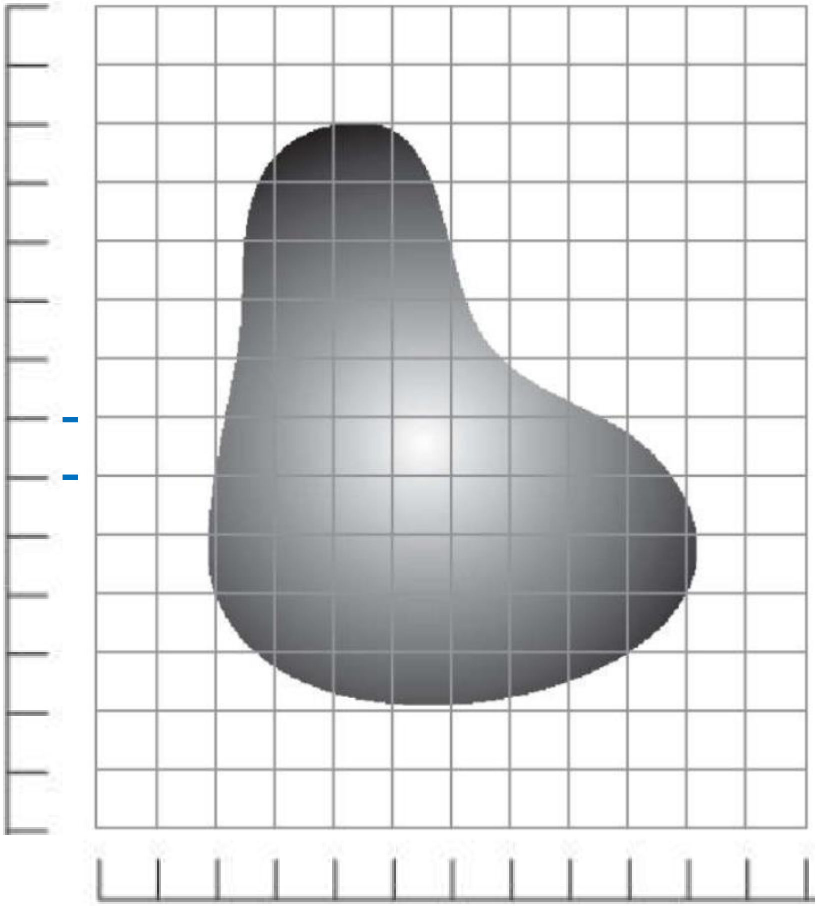
Basic BIP Concepts

Physical signal (continous signal) → Digitalization
- Sampling
- Quantization

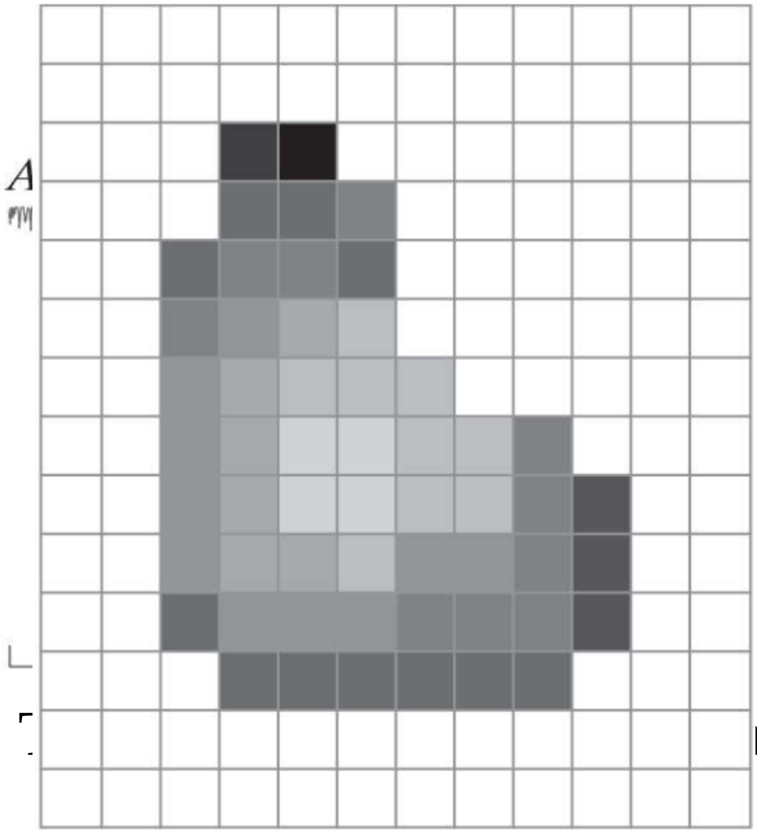


Suppose: we have a signal | Temporal sampling | Digitalization | Digital Signal

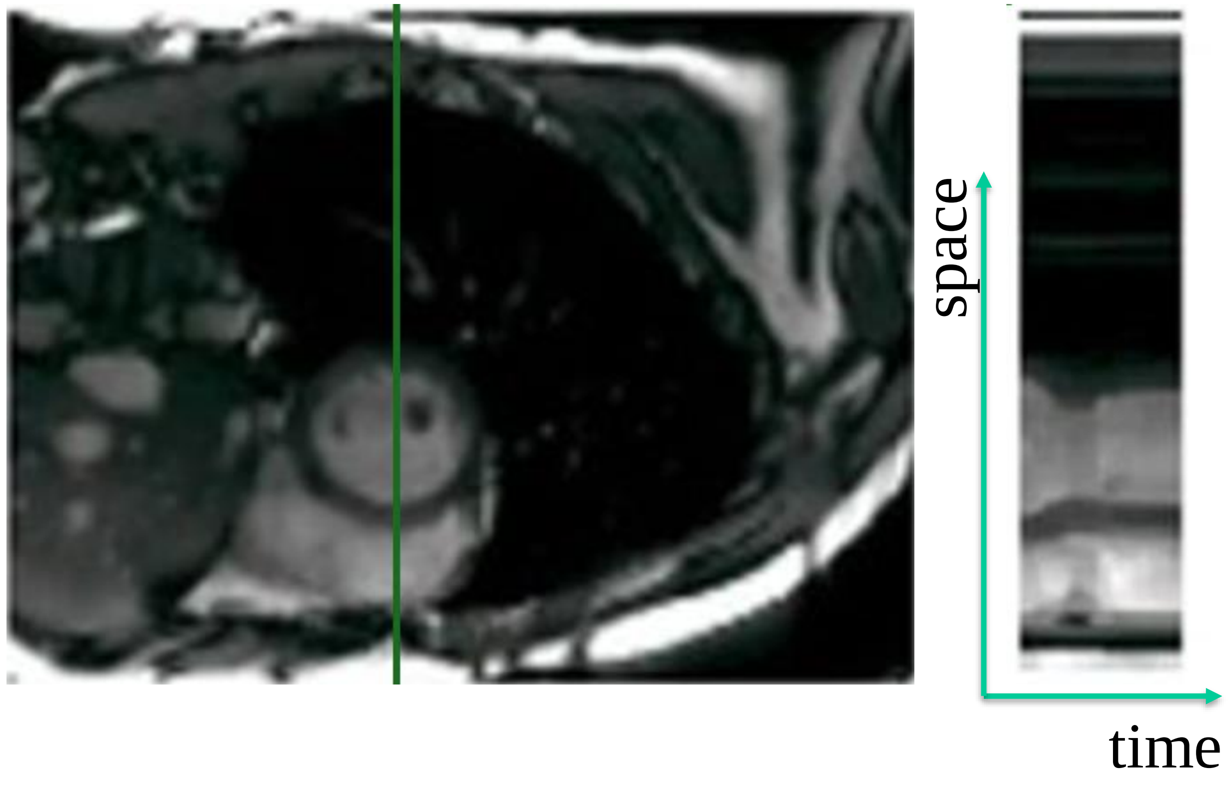
Sampling



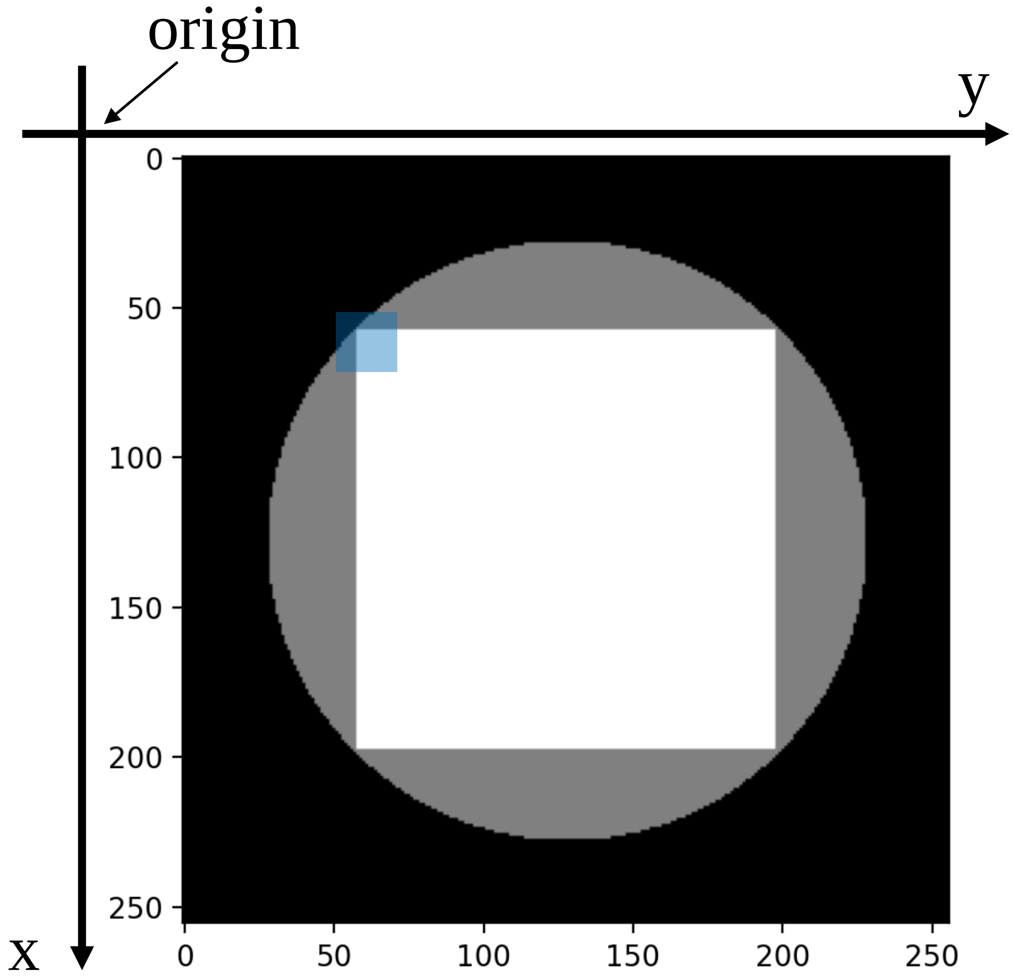
Line plot



Line plot example

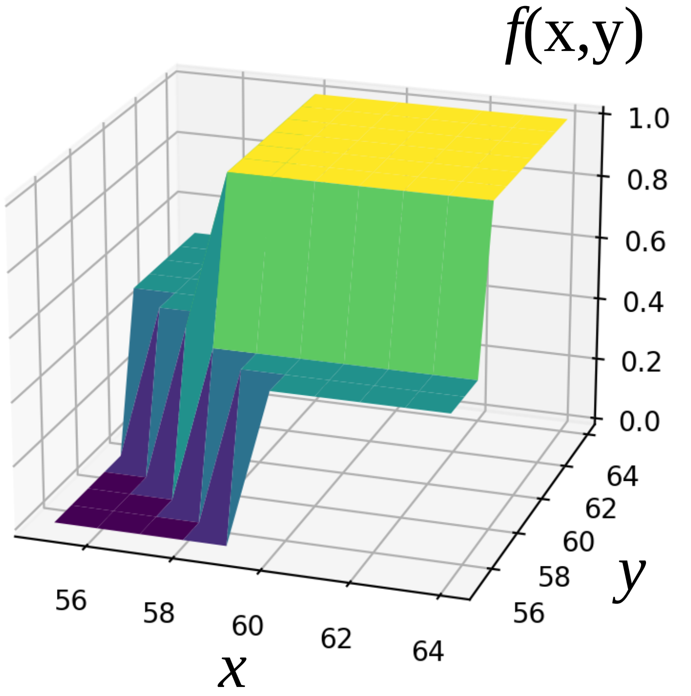


Sampling



(x,y)=([55-64], [55-64])

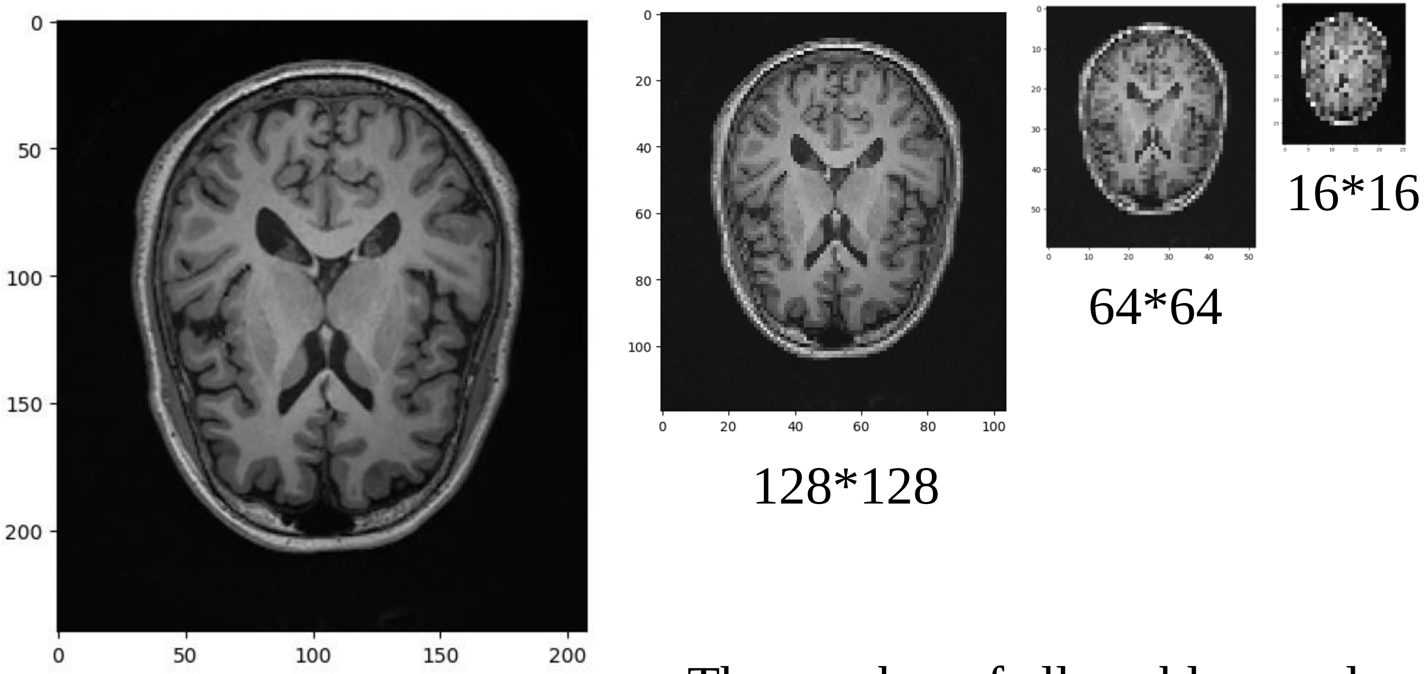
0	0	0	0	0	.5	.5	.5	.5	.5
0	0	0	0	.5	.5	.5	.5	.5	.5
0	0	0	.5	.5	.5	.5	.5	.5	.5
0	0	.5	1	1	1	1	1	1	1
0	.5	.5	1	1	1	1	1	1	1
.5	.5	.5	1	1	1	1	1	1	1
.5	.5	.5	1	1	1	1	1	1	1
.5	.5	.5	1	1	1	1	1	1	1
.5	.5	.5	1	1	1	1	1	1	1
.5	.5	.5	1	1	1	1	1	1	1



Sampling

$x \in Z$
$y \in Z$ $\quad f(x,y)$

$$A = \begin{bmatrix} f(0,0) & \cdots & f(0,N) \\ \vdots & \ddots & \vdots \\ f(M,0) & \cdots & f(M,N) \end{bmatrix}$$

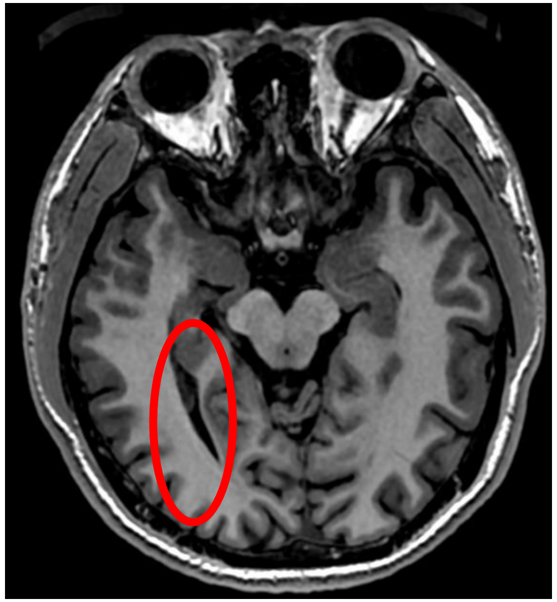


256*256 128*128 64*64 16*16

The number of allowable gray levels was kept the same.

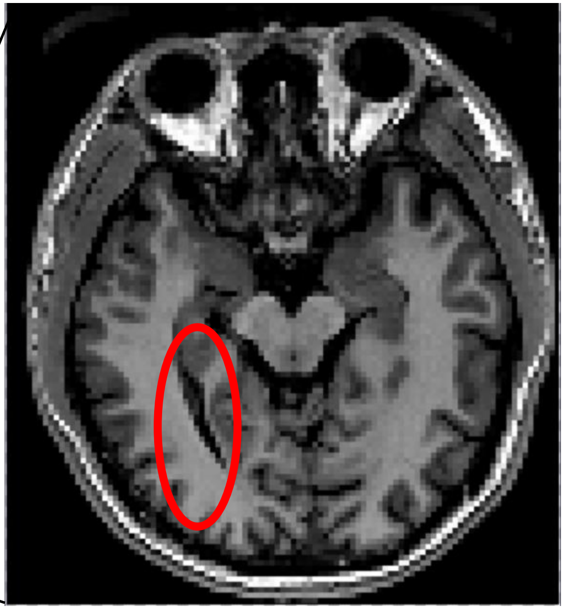
Sampling

Matrix is (256,256)

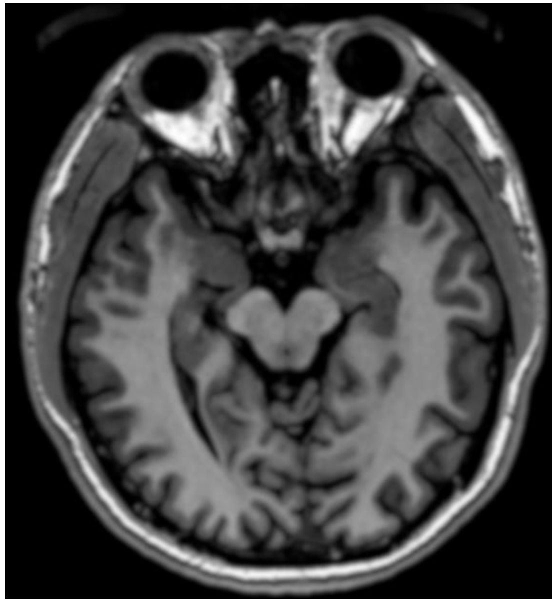


(128,128)

(128,128)



(256,256)



Resolution=1x1mm

Resolution=2x2mm

Resolution?

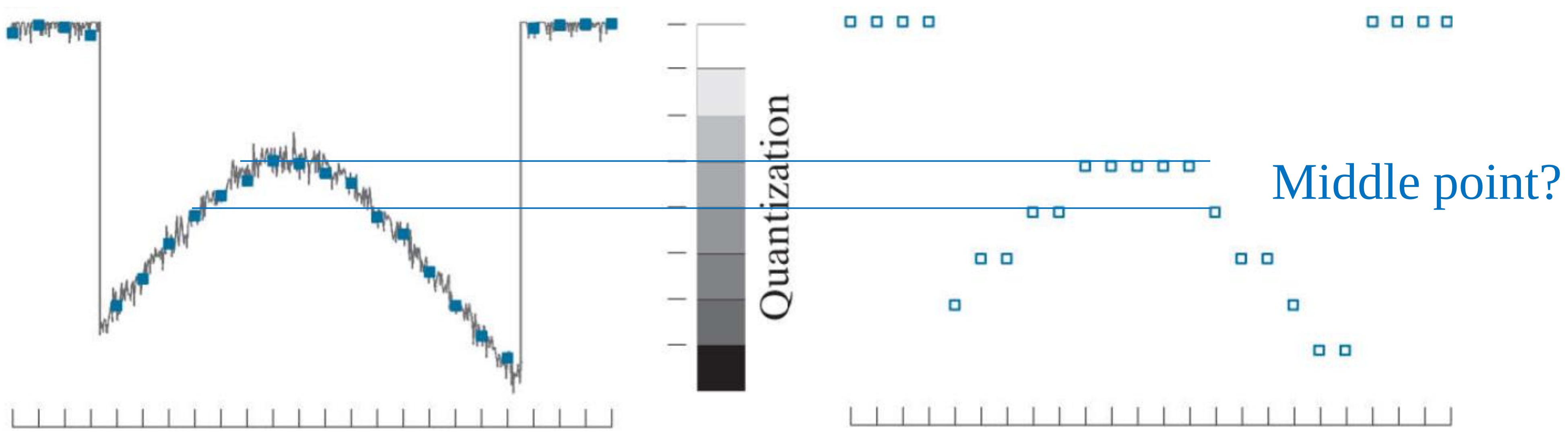
Physical resolution is NOT matrix size only!

Sampling Summary

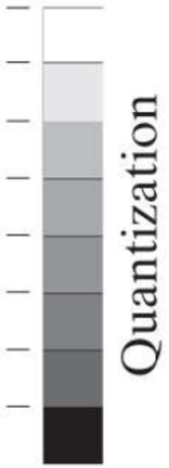
- Spatial Image → Matrix Index
- Physical resolution vs matrix size

- Questions?

Quantization

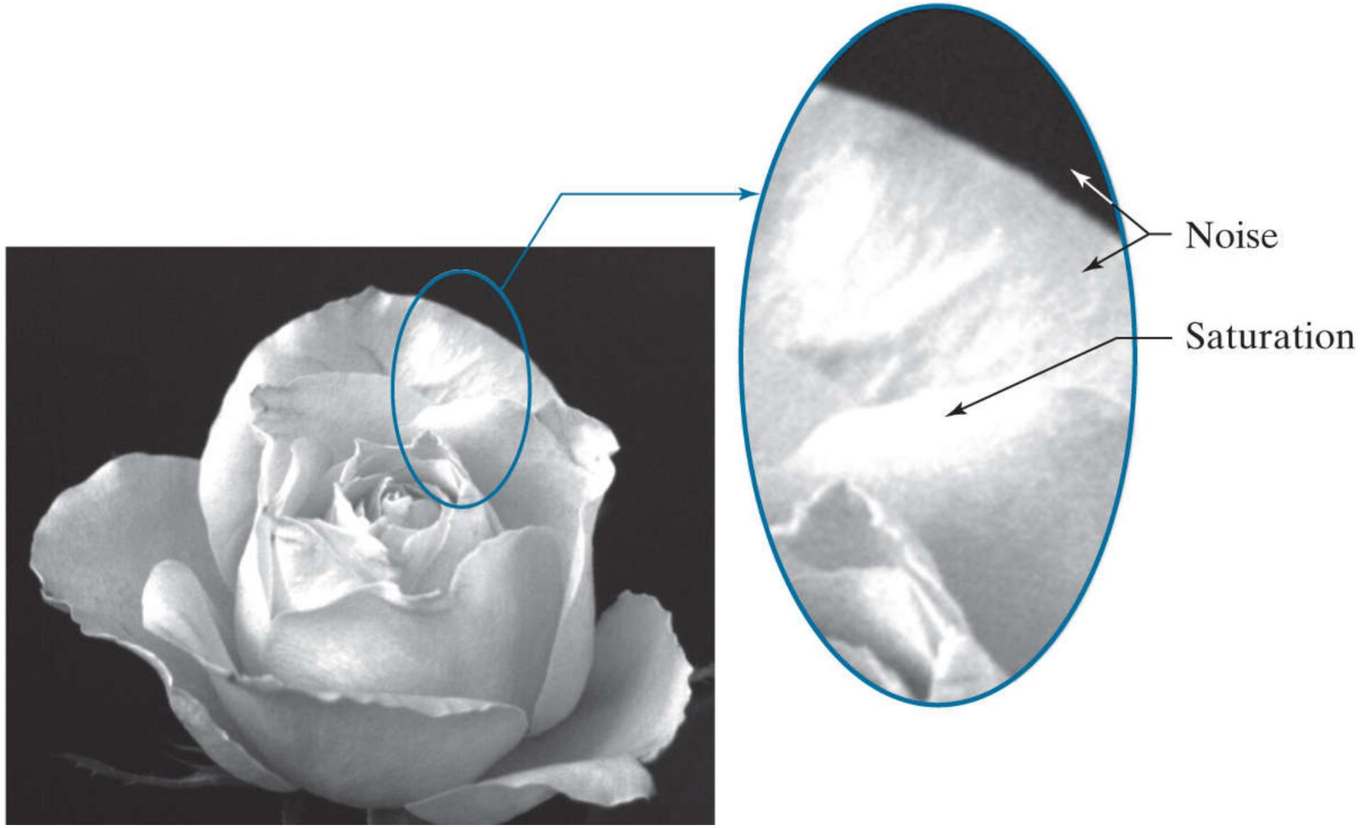


Quantization



$$L = 2^K$$

K=8 [0-255]



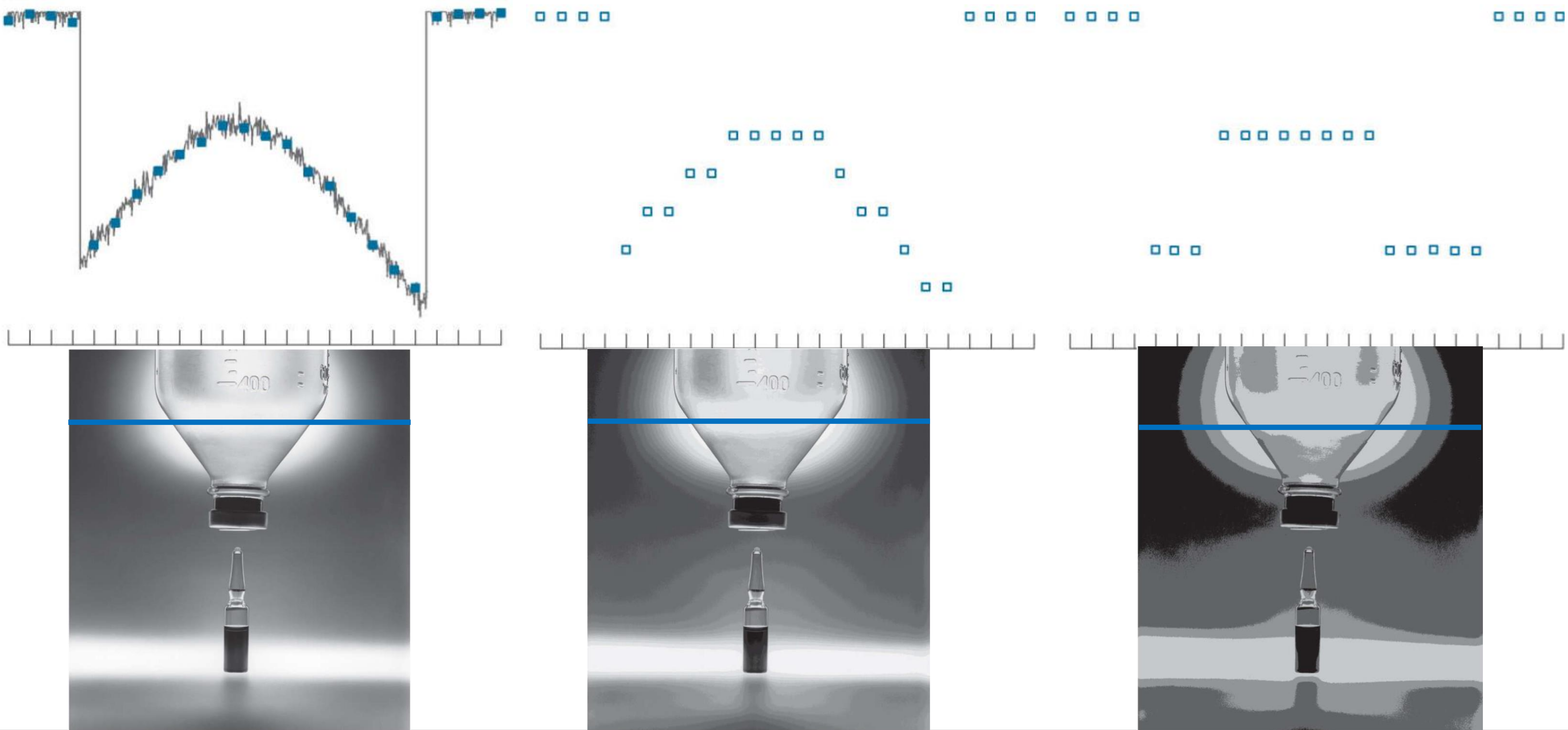
Dynamic range

Storage size = M*N*K

Quantization

K=3 [0-7]

K=2 [0-3]

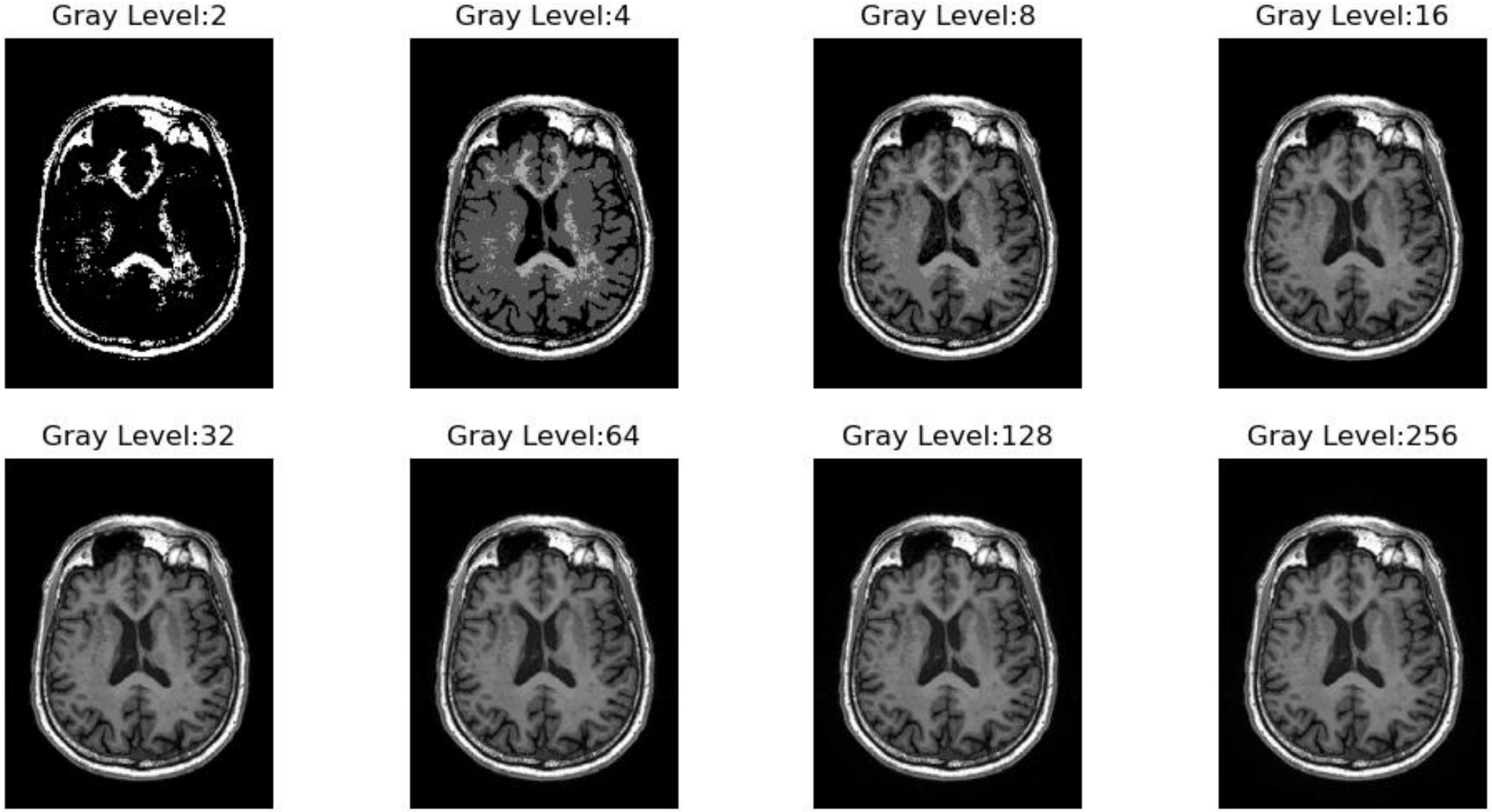


Quantization

High Dynamic Range (HDR)

Quantization

- Pixel intensity determined by bit depth: 2^N levels (e.g., 8-bit=256 levels)
- Range: 0(Black) ~ 2^N-1(White), representing brightness intensity.



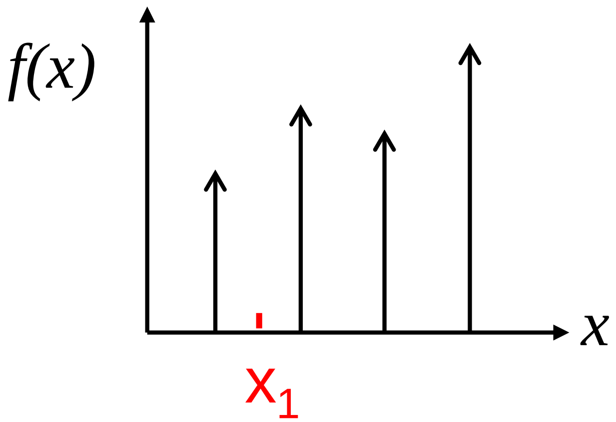
Quantization Summary

- Image Intensity → Matrix value
- Image range: K

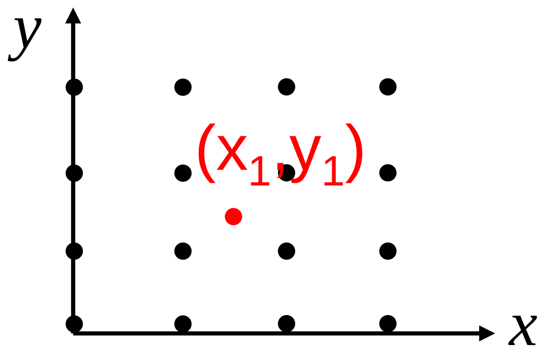
- Questions?

Interpolation

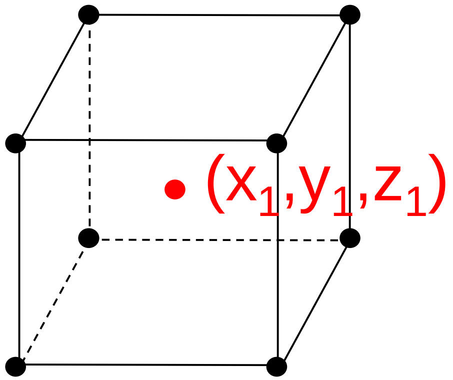
- **Interpolation is a basic tool used extensively in tasks such as zooming, shrinking, rotation, and other transformations**
- **The model-based recovery of continuous data from discrete data within a known range of abscissas**



1-D



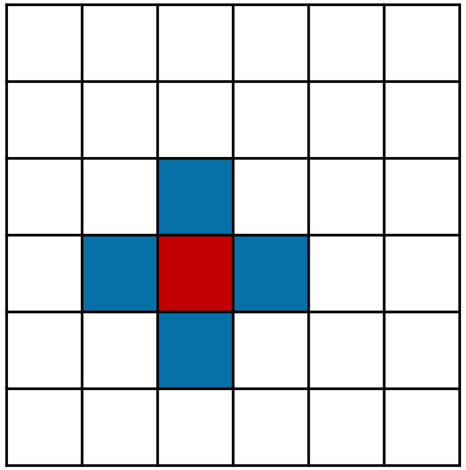
2-D



3-D

Pixel Neighborhood

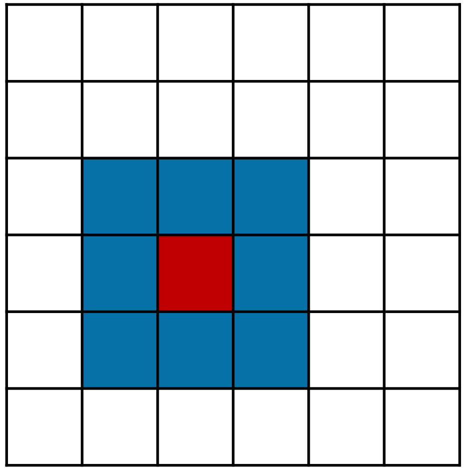
- Neighbors
  - 4-neighbors
  - $N_4(p)$



(x,y)

(x+1,y),(x-1,y),(x,y+1),(x,y-1)

  - 8-neighbors
  - $N_8(p)$

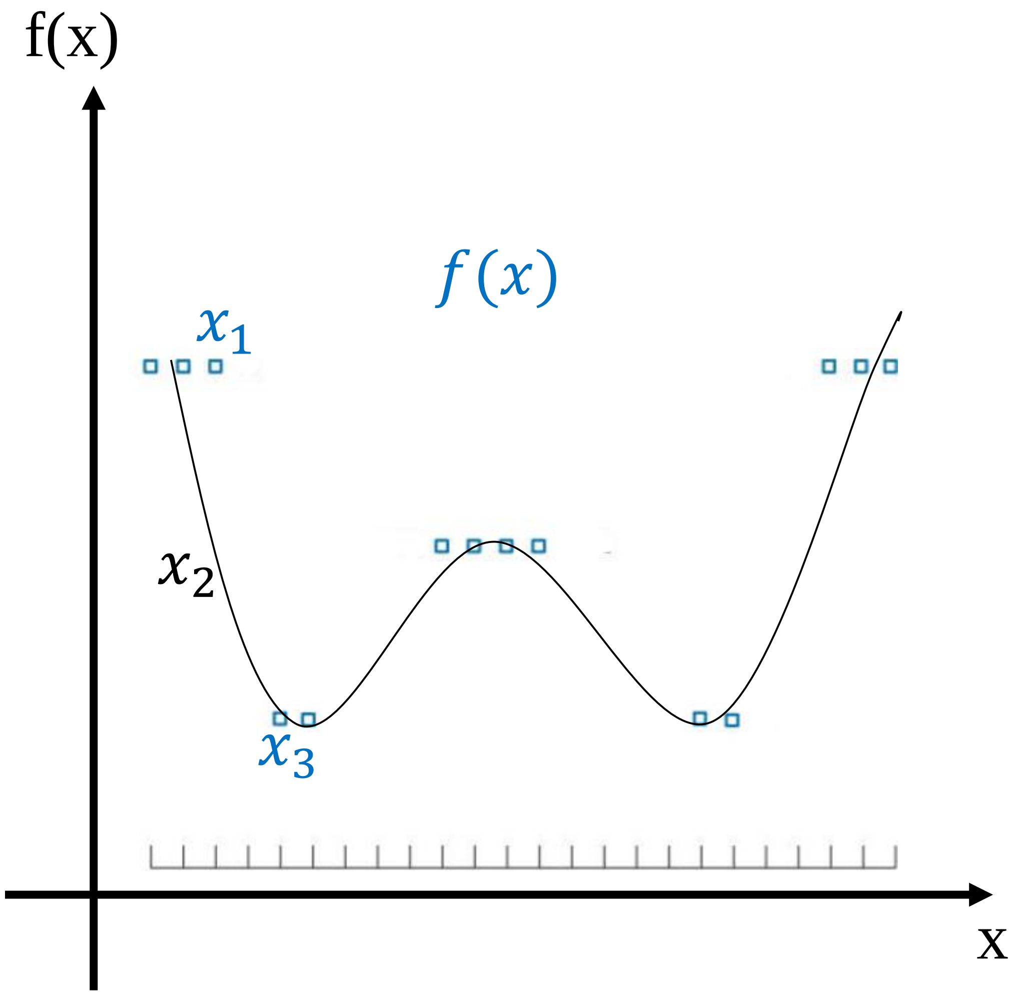


(x+1,y),(x-1,y),(x,y+1),(x,y-1)

(x+1,y+1),(x-1,y+1),(x-1,y+1),(x-1,y-1)

$N_4(p) + N_D(p)$=$N_8(p)$

Interpolation (1D)



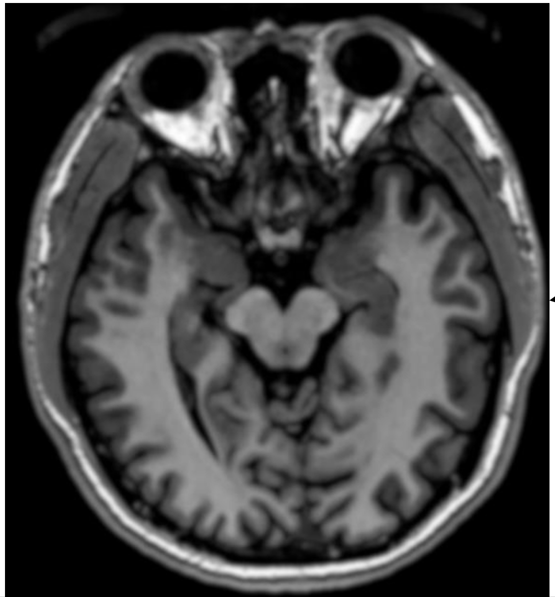
$$f(x_i) = g(f(x_{i-1}), f(x_{i+1}))$$

e.g. $f(x_2) = \frac{f(x_1)+f(x_3)}{2}$ linear

e.g. $f(x_2) = f(x_1)$ nearest

Interpolation (2D)

Interpolation



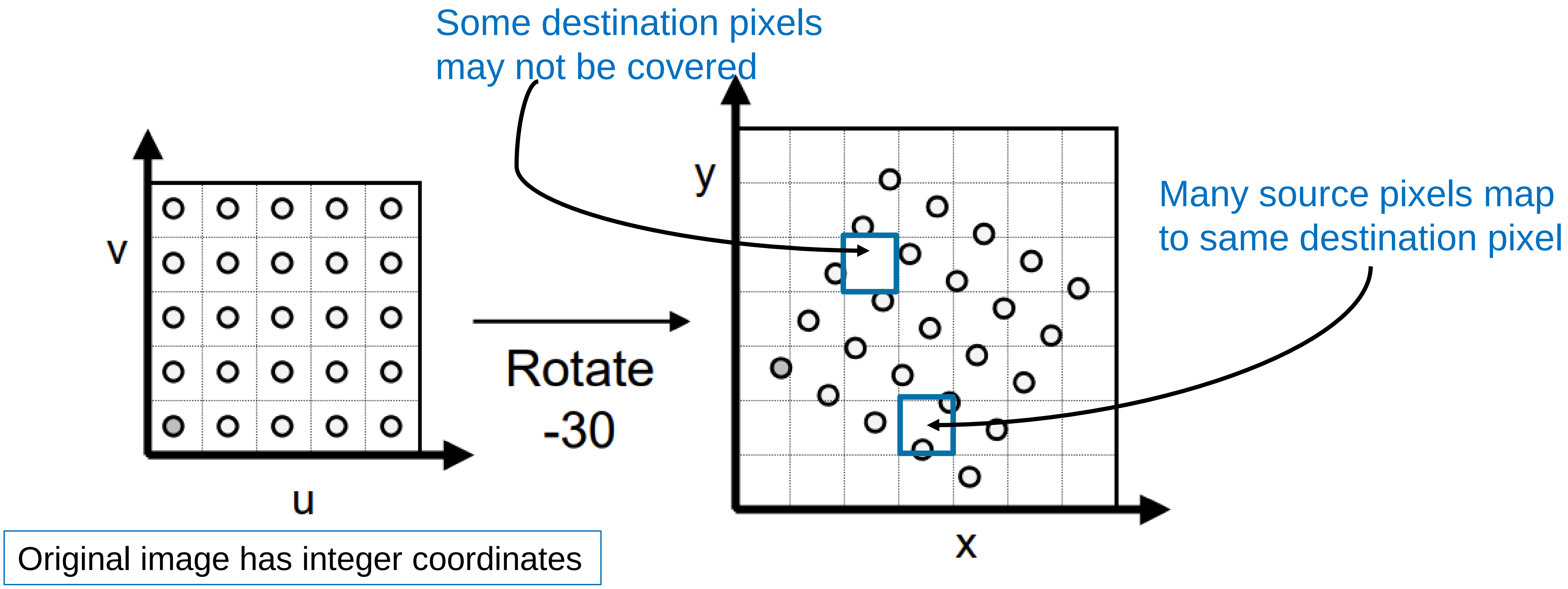
e.g. $f(x,y) = ax + by + cxy + d$ bilinear

e.g. $f(x,y) = \sum_{i=0}^{3}\sum_{j=0}^{3} a_{ij}x^i y^j$ bicubic

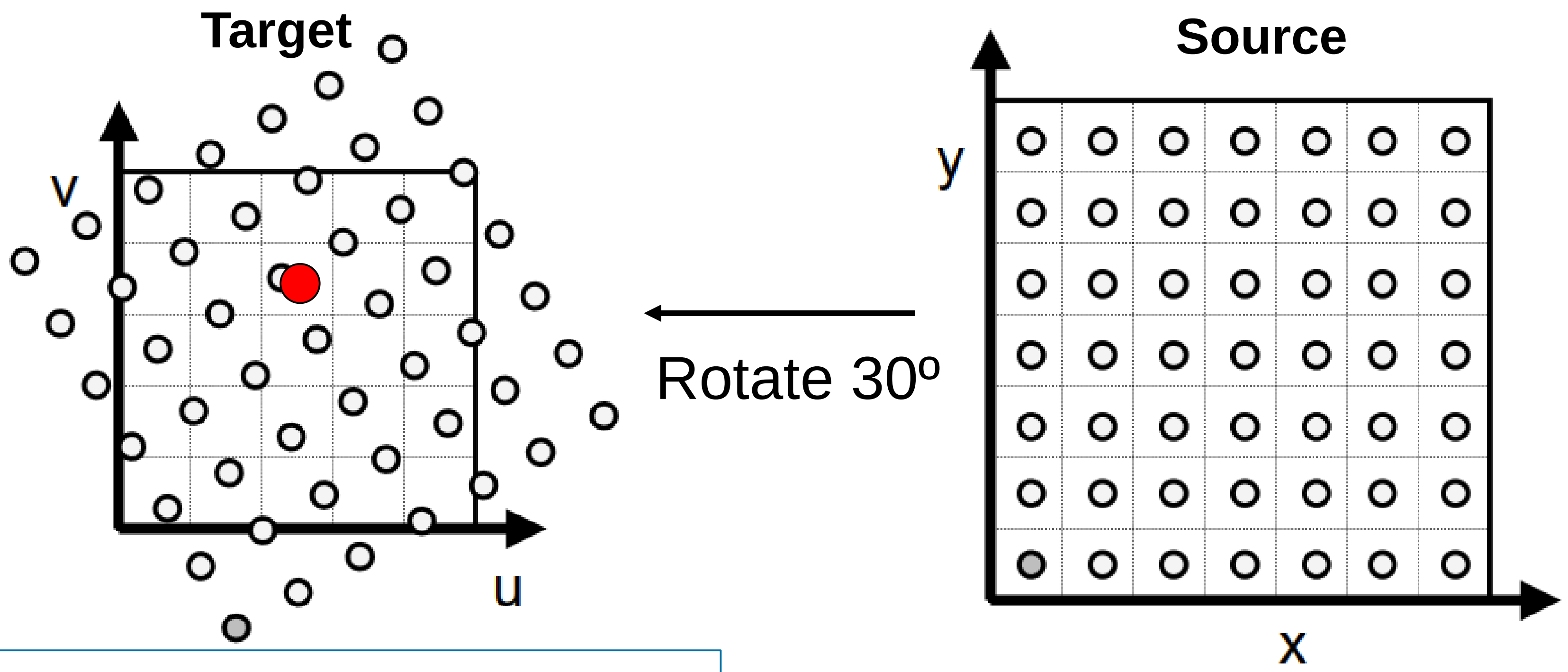
- a, b, c, d is obtained from the N4 neighborhood
- a_{ij} is obtained from N16 neighborhood

Spline, wavelet, deep learning (super-resolution)…

Why interpolation?



Why interpolation?

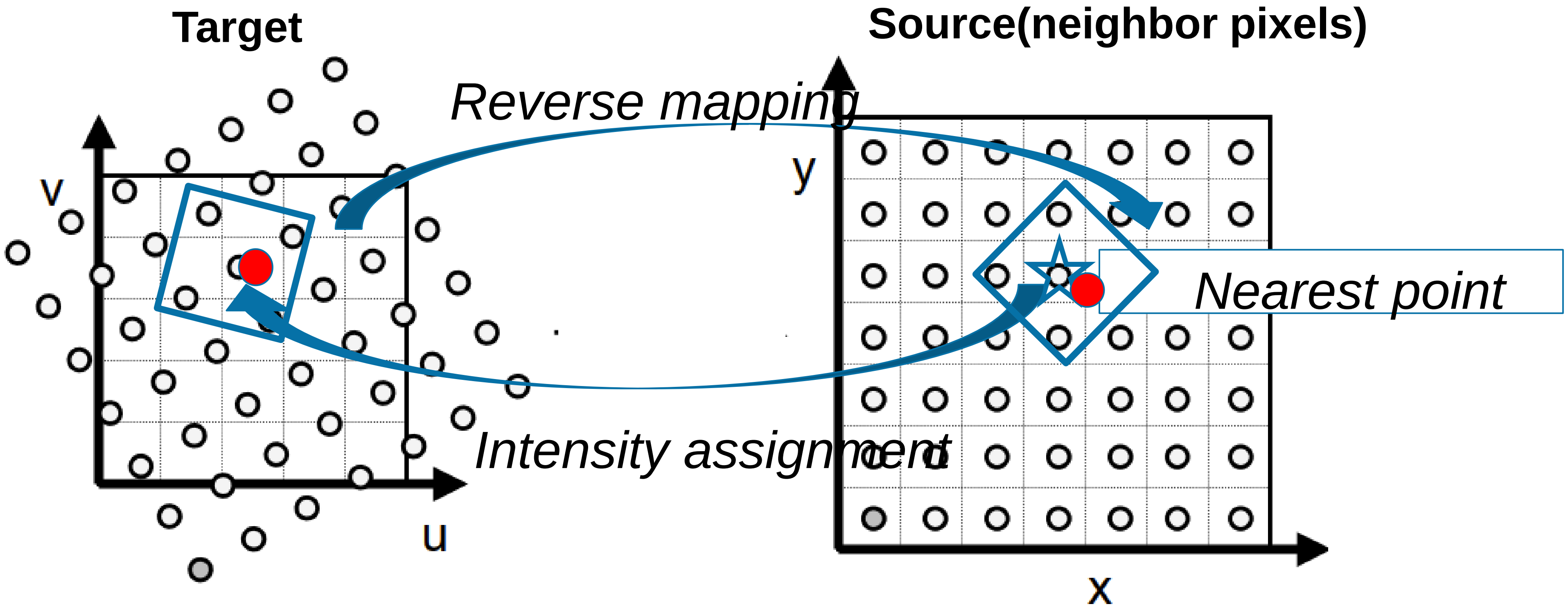


Transformed image usually does not have integer coordinates

must resample source!

Interpolation methods

- **Nearest interpolation**
- Bilinear interpolation
- Bi-Cubic interpolation

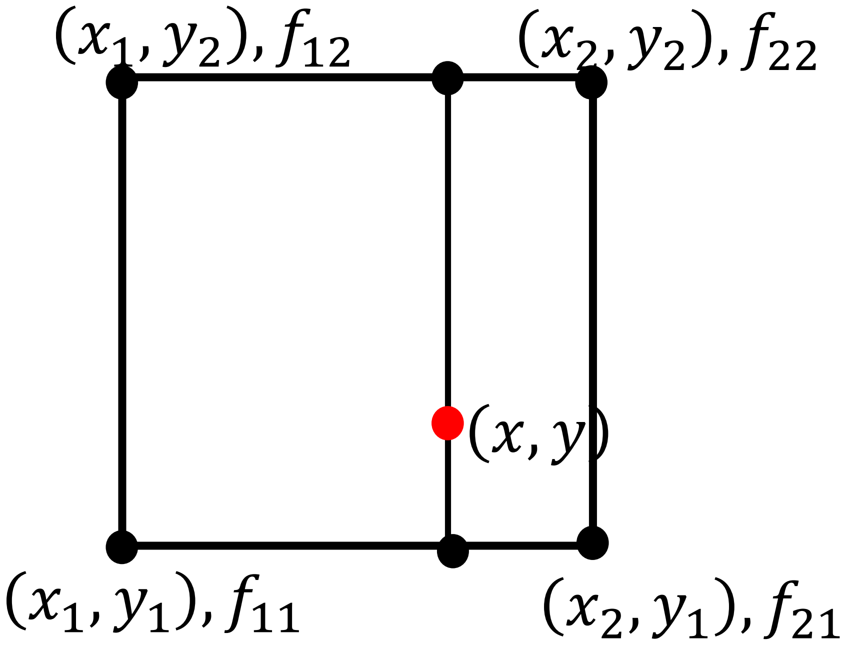


Interpolation methods

- Nearest interpolation
- **Bilinear interpolation**
- Bi-Cubic interpolation

$$f(x, y) = ax + by + cxy + d$$

2D: Source matrix (4 pixels)



Interpolation methods

- Nearest interpolation
- **Bilinear interpolation**
- Bi-Cubic interpolation

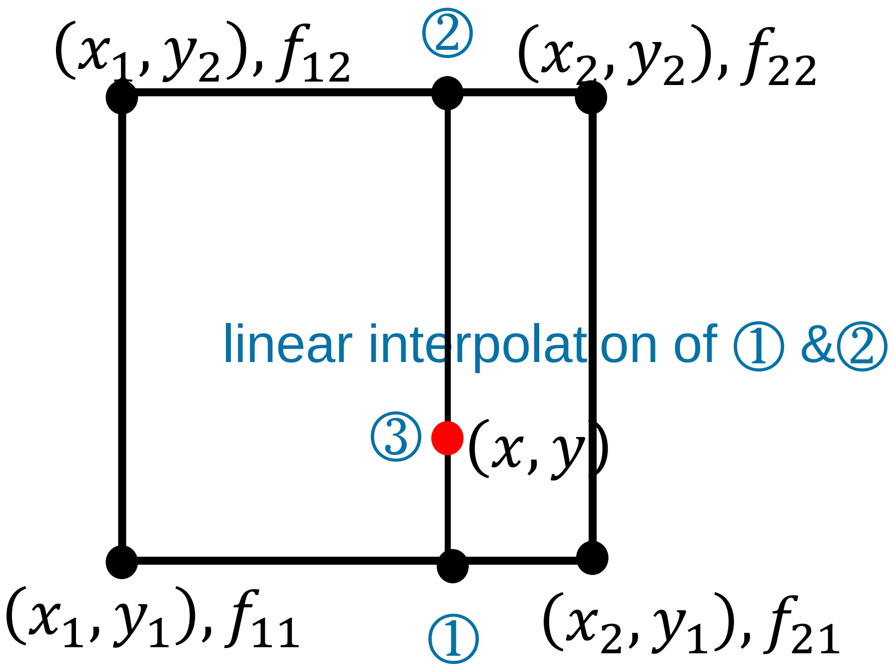
$$f(x, y) = ax + by + cxy + d$$

① R1 = $(x - x_1)(f_{21} - f_{11}) / (x_2 - x_1) + f_{11}$

② R2 = $(x - x_1)(f_{22} - f_{12}) / (x_2 - x_1) + f_{12}$

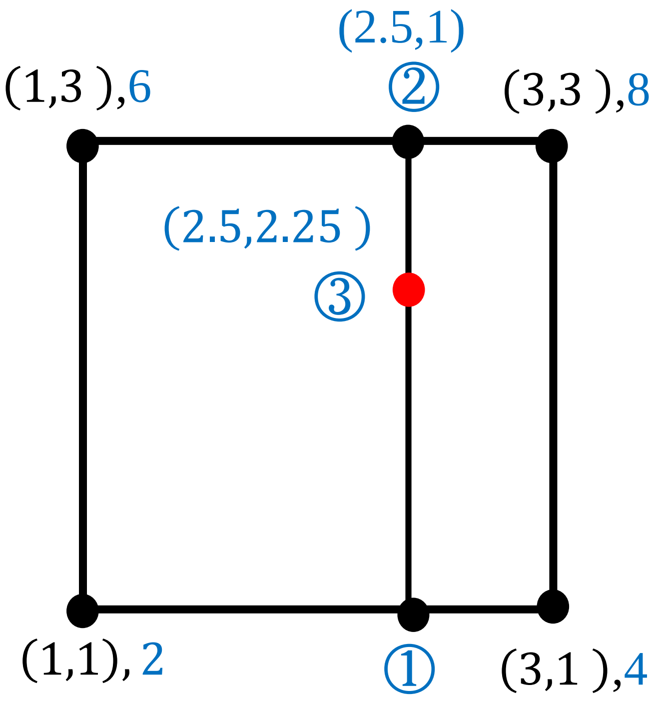
③ $f(\mathrm{x}, \mathrm{y}) = (\mathrm{y} - \mathrm{y}_1) * (\mathrm{R2} - \mathrm{R1}) / (\mathrm{y}_2 - \mathrm{y}_1) + \mathrm{R1}$

2D: Source matrix (4 pixels)



Interpolation methods

- Nearest interpolation
- Bilinear interpolation
- Bi-Cubic interpolation



Steps:

① R1 = (2.5 – 1)(4 – 2) / (3-1)+2=3.5

② R2 = (2.5 – 1)(8 – 6) / (3-1)+6=7.5

③ f(x, y) = (2.25-1) * (7.5-3.5) / (3-1) + 3.5 = 6

Interpolation methods

- Nearest interpolation
- Bilinear interpolation
- Bi-Cubic interpolation
- ……

1-D (4 pixels):

$$f(x) = \sum_{i=0}^{3} a_i x^i$$

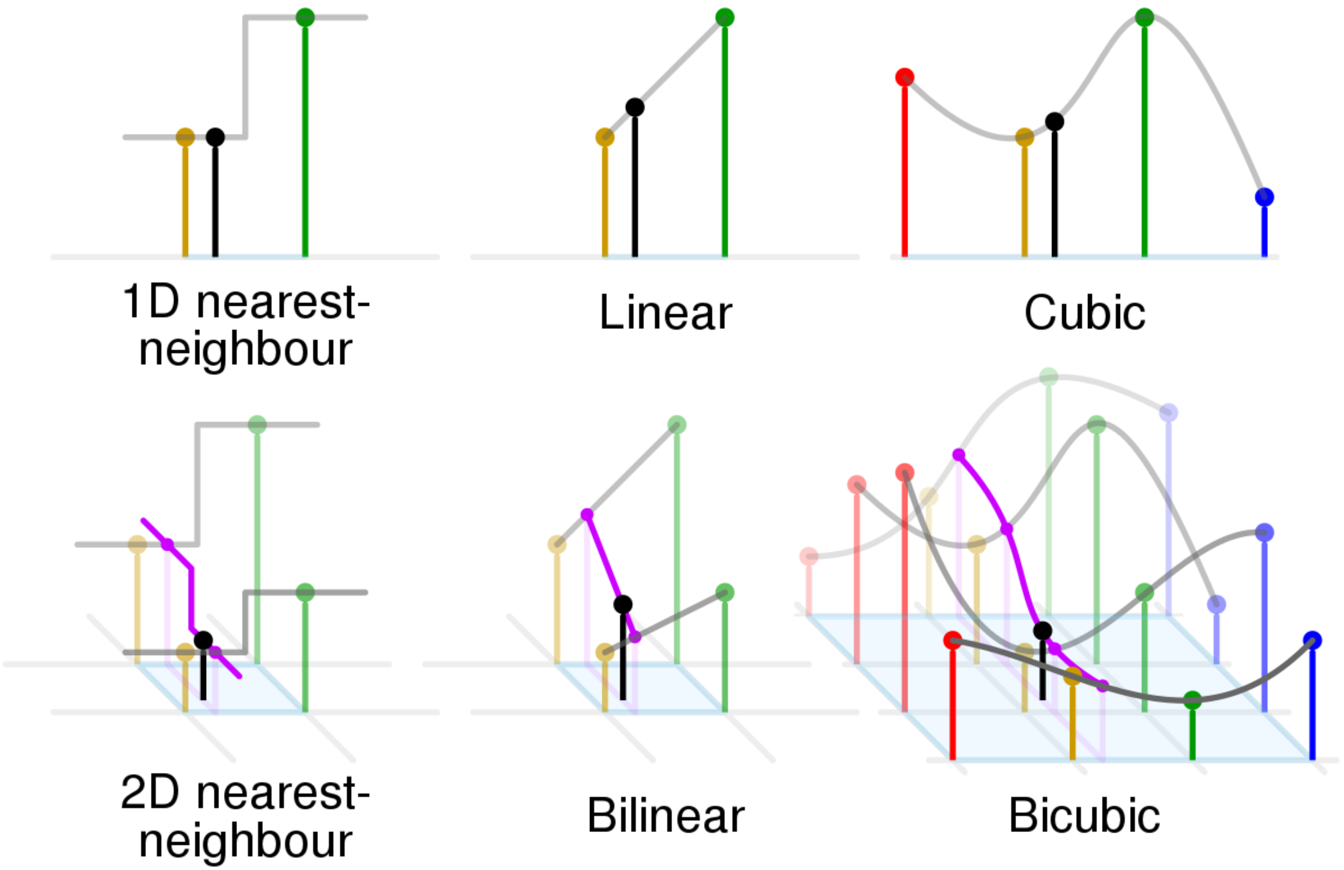
- $y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$
- Satisfy: $y(x_i) = y_i$,
  $y(x_{i+1}) = y_{i+1}$
- Lagrange basis function: $L_i(x) = \prod_{j=0, j\neq i}^{n} \frac{x - x_j}{x_i - x_j}$
- $y(x) = \sum_{i=0}^{n} y_i \, L_i(x)$

Interpolation methods

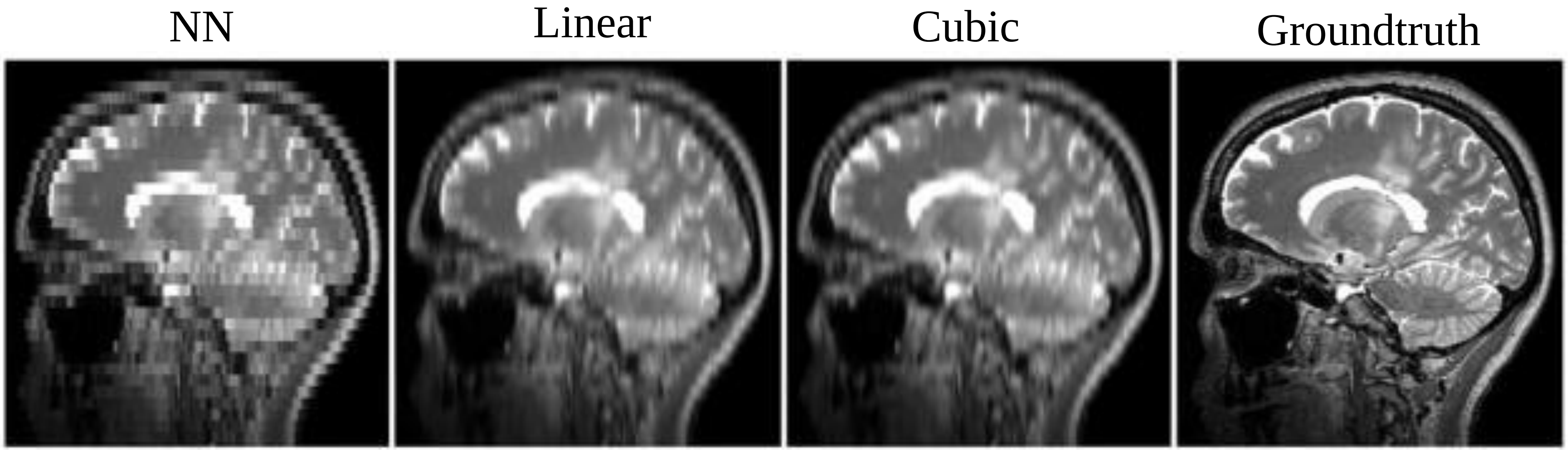
- Nearest interpolation
- Bilinear interpolation
- Bi-Cubic interpolation
- ……

2-D (16 pixels):

$$f(x,y) = \sum_{i=0}^{3}\sum_{j=0}^{3} a_{ij}x^i y^j$$



Interpolation

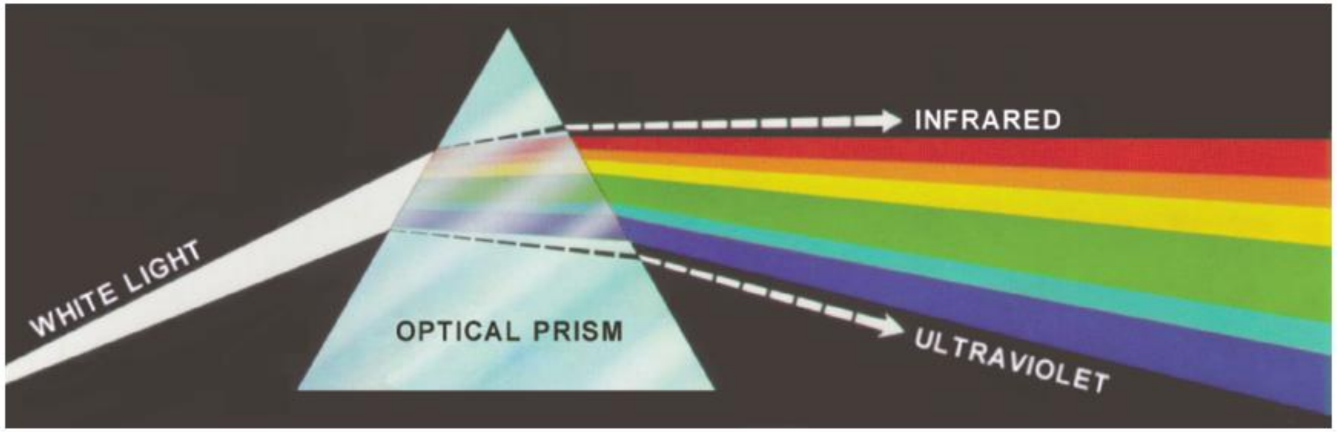


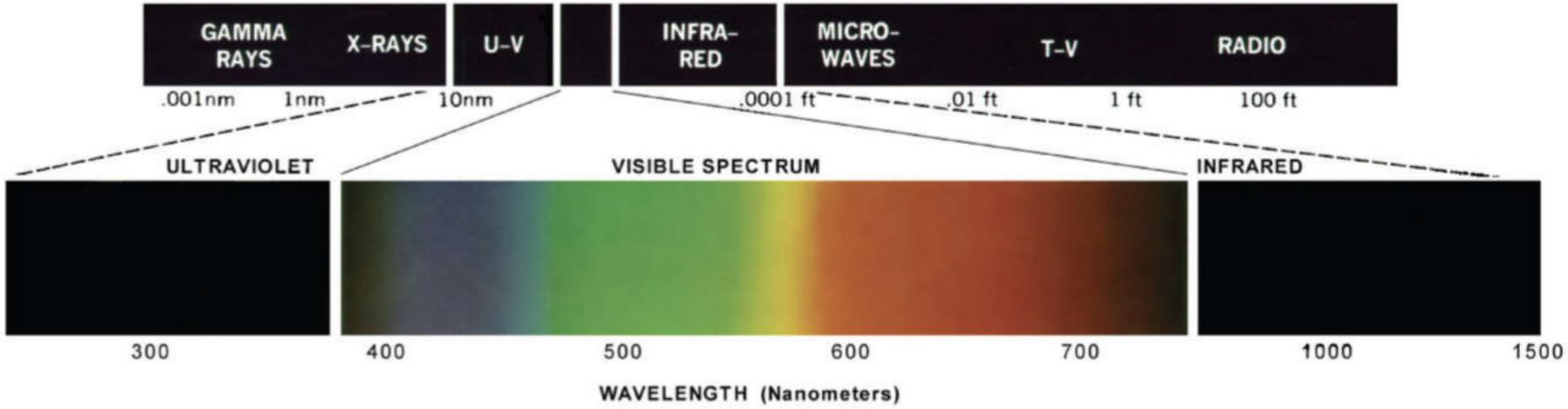
Summary

- Sampling: Spatialize Image → Matrix Index → N*M
- Quantization: Image intensity → Matrix Value → K
- Interpolation: Index (x,y) + intensity (v)

Color Images

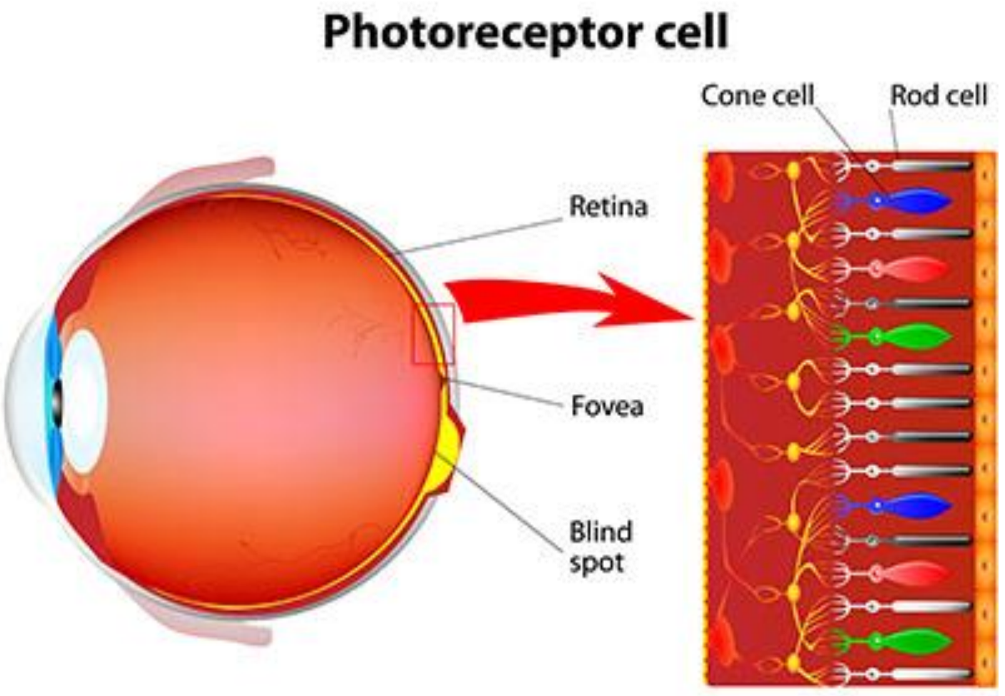
Color Fundamentals





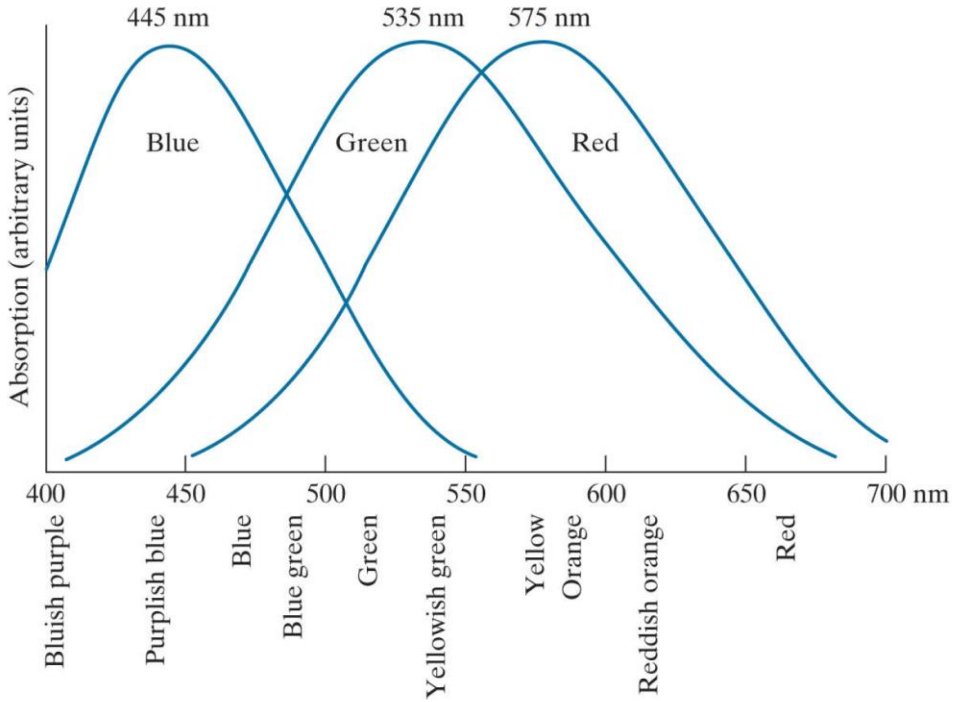
Sensor in the Eyes

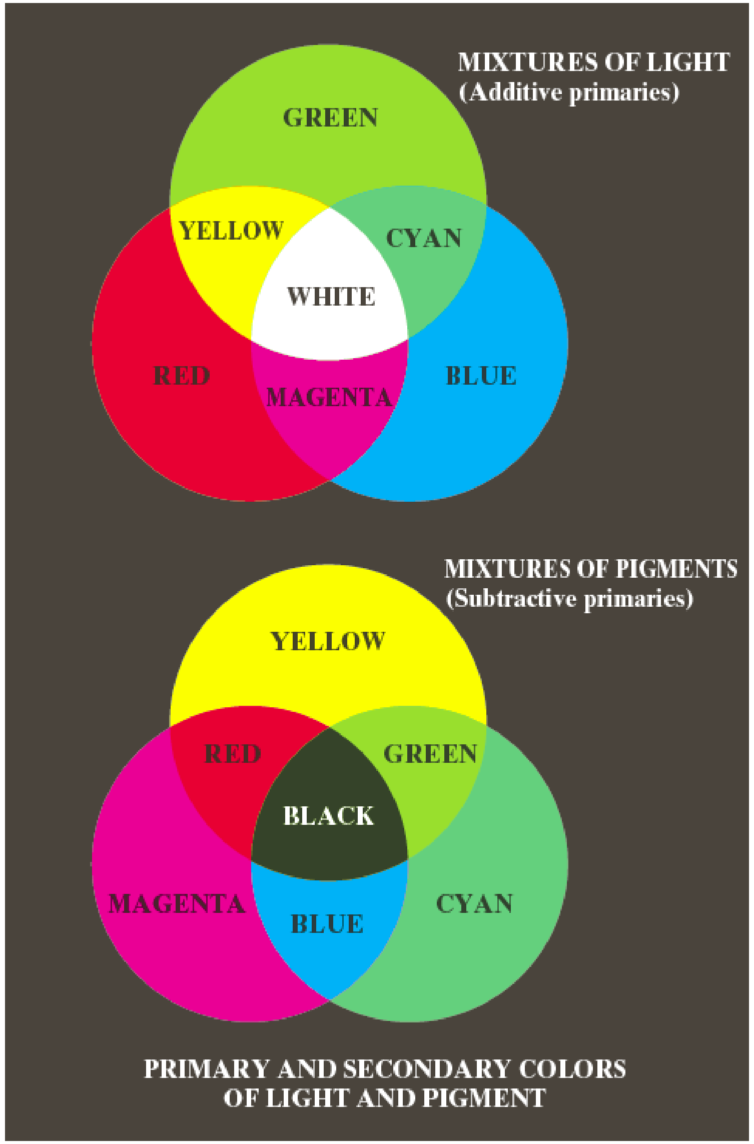
- Cones: sensors in the eyes responsible for color vision.
  - 6-7 million cones in human eye
  - 65% of all cones are sensitive to red light, 33% to green, 2% to blue but the most sensitive



Color Fundamentals

- Thus, primary colors are RGB
  - Spectrum distribution
  - Can NOT make all the colors



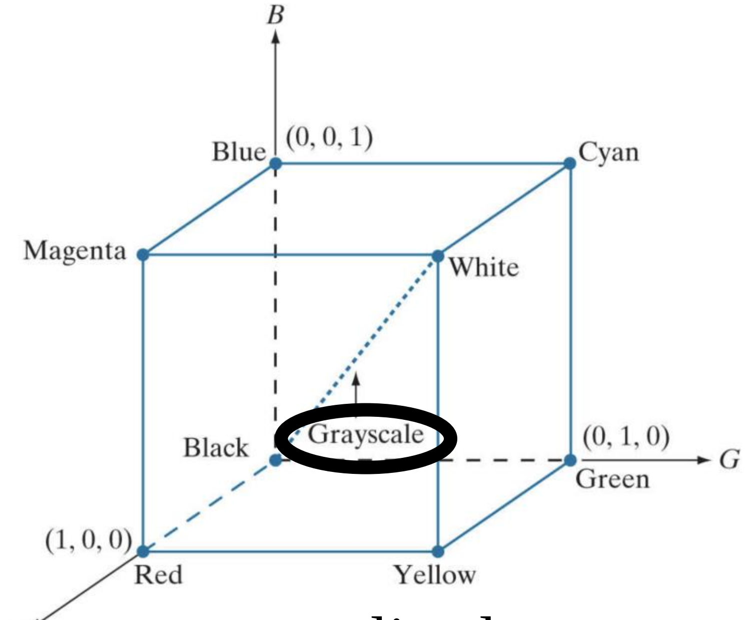


Reflect

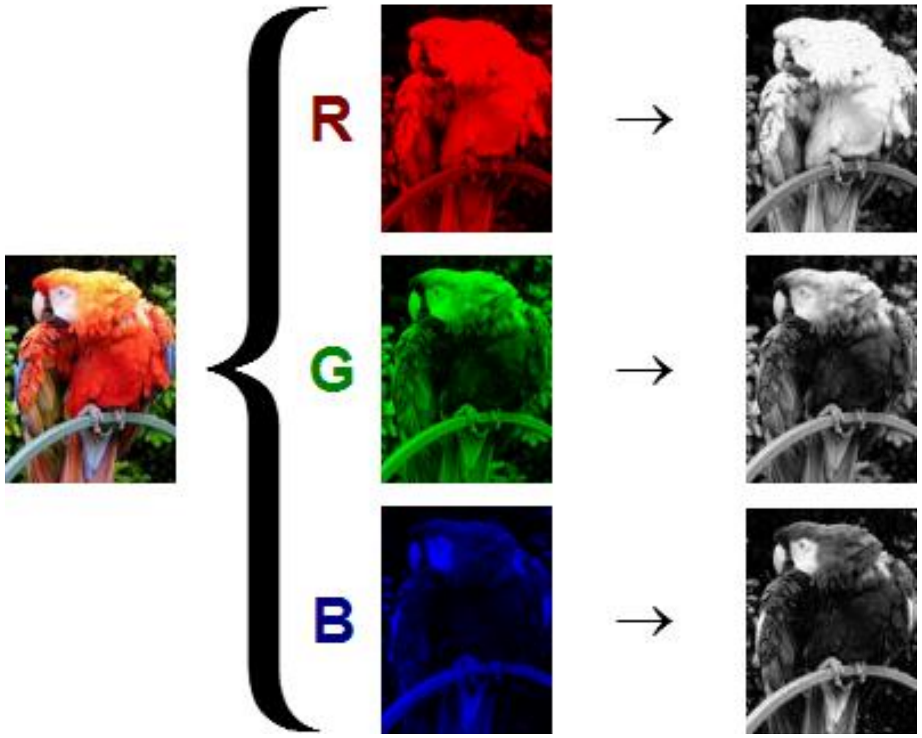
Absorption

Color models

- Color models based on hardwares
  - E.g. RGB for the monitor or camera



normalized



Pixel depth: the number of bits to represent the color

8 bits = [0,255]→$(2^8)^3$=16777216

Color models

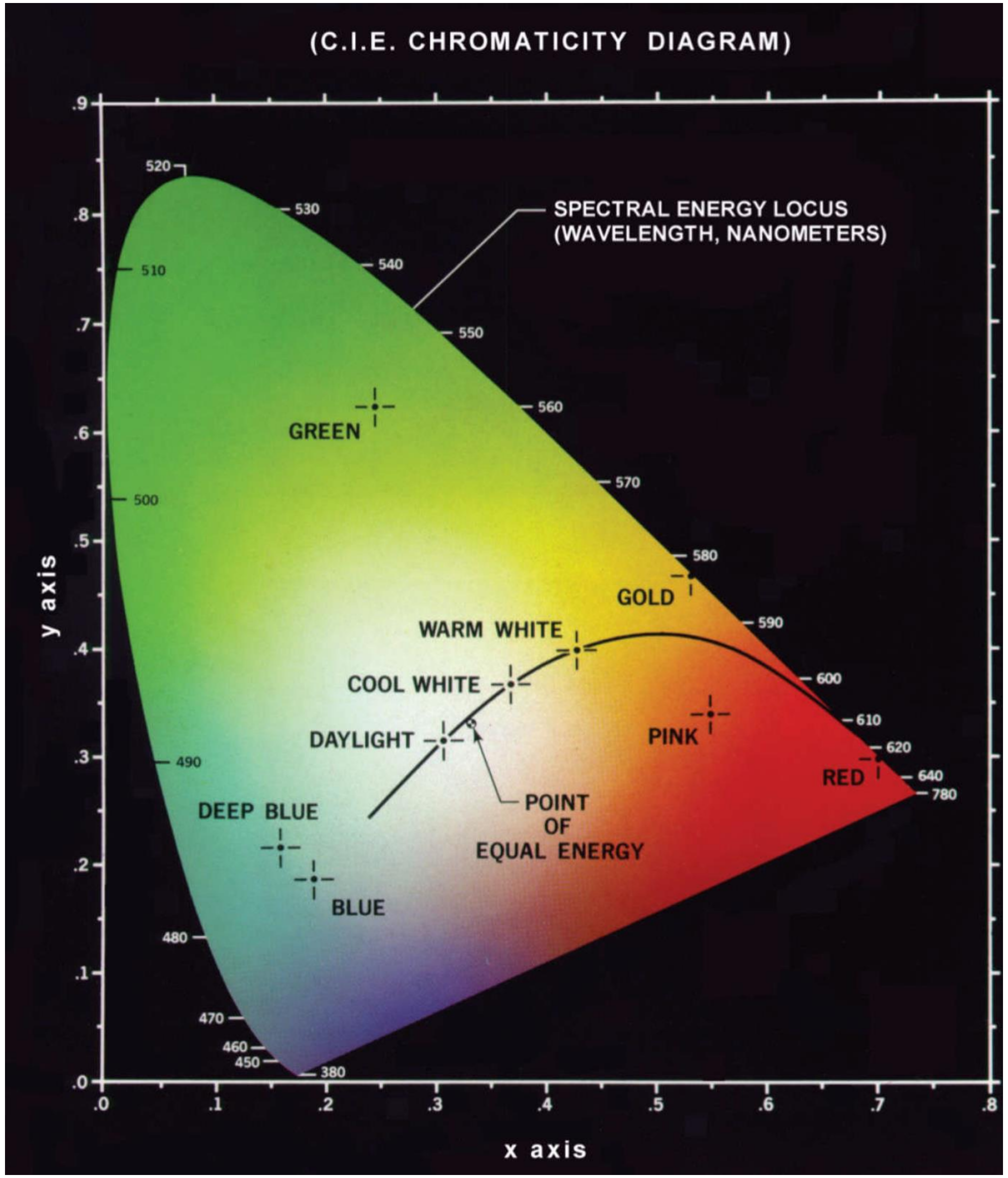
- CMY for the pigments and printer

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = 1 - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- CMYK
  - K: true black

Color models

- HIS for human eyes
  - Hue: mixed wavelength, perceived by an observer
  - Saturation: purity, amount of white light in the hue
    - Pink = a · red + b · white
  - Intensity: brightness (colorless)



Color models

- From RGB to HSI

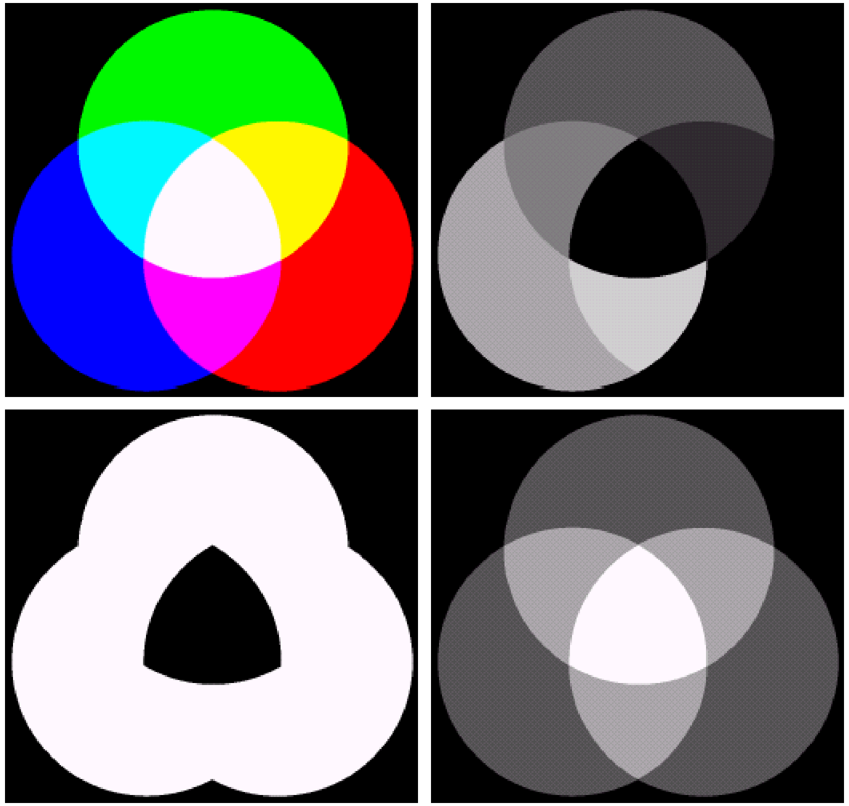
$$I = \frac{R+G+B}{3}$$

$$S = 1 - \frac{3}{R+G+B}\min(R,G,B)$$

$$\theta = \cos^{-1}\left\{\frac{0.5[(R-G)+(R-B)]}{[(R-G)^2+(R-B)(G-B)]^{1/2}}\right\} \qquad H = \begin{cases} \theta & \text{if } B \le G \\ 360-\theta & \text{if } B > G \end{cases}$$

Color models

- Example: RGB → HSI



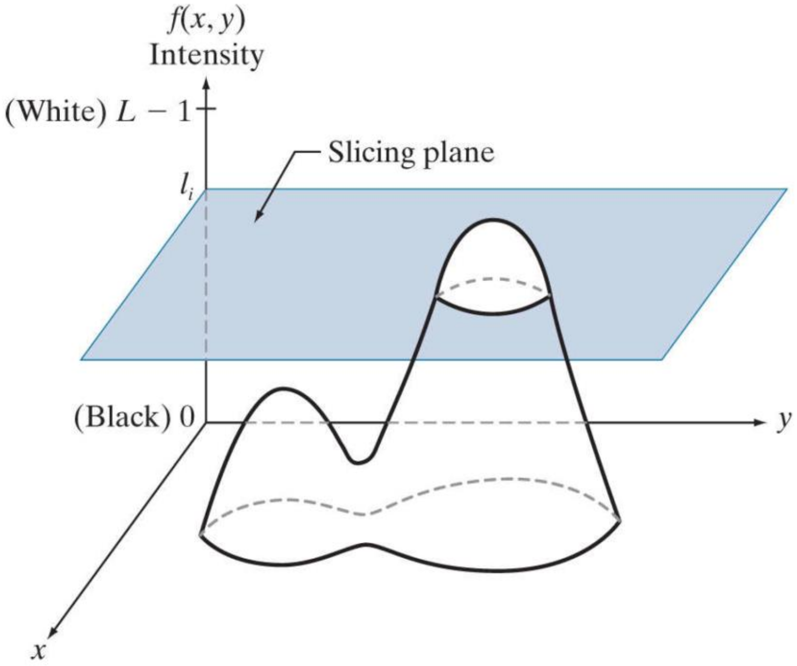
hue

saturation

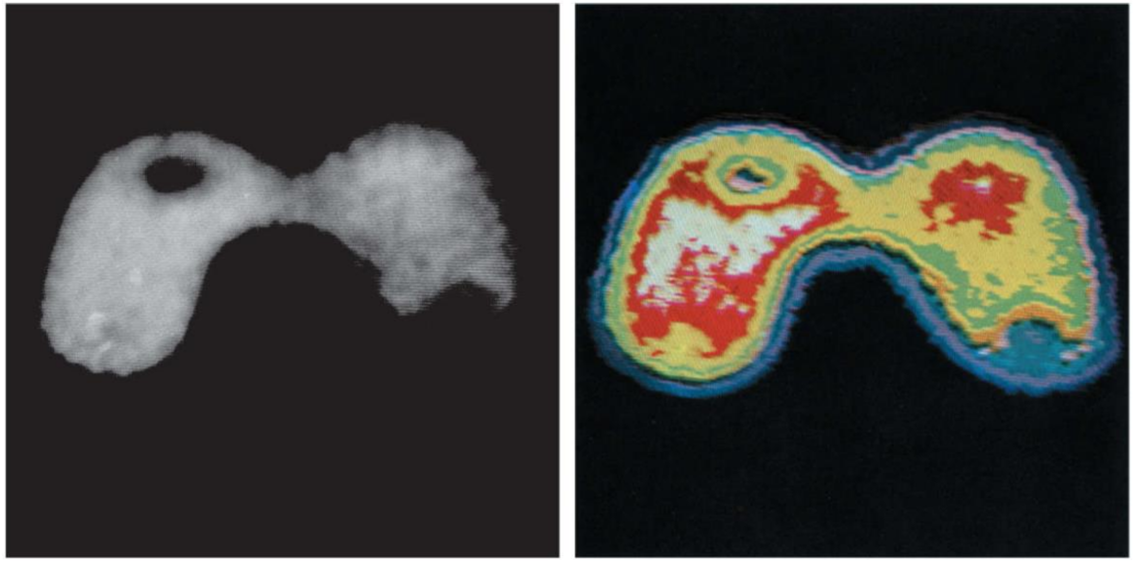
intensity

Pseudocolor

- False color
  - Humans can discern color shades and intensities more
  - Intensity slicing and color coding



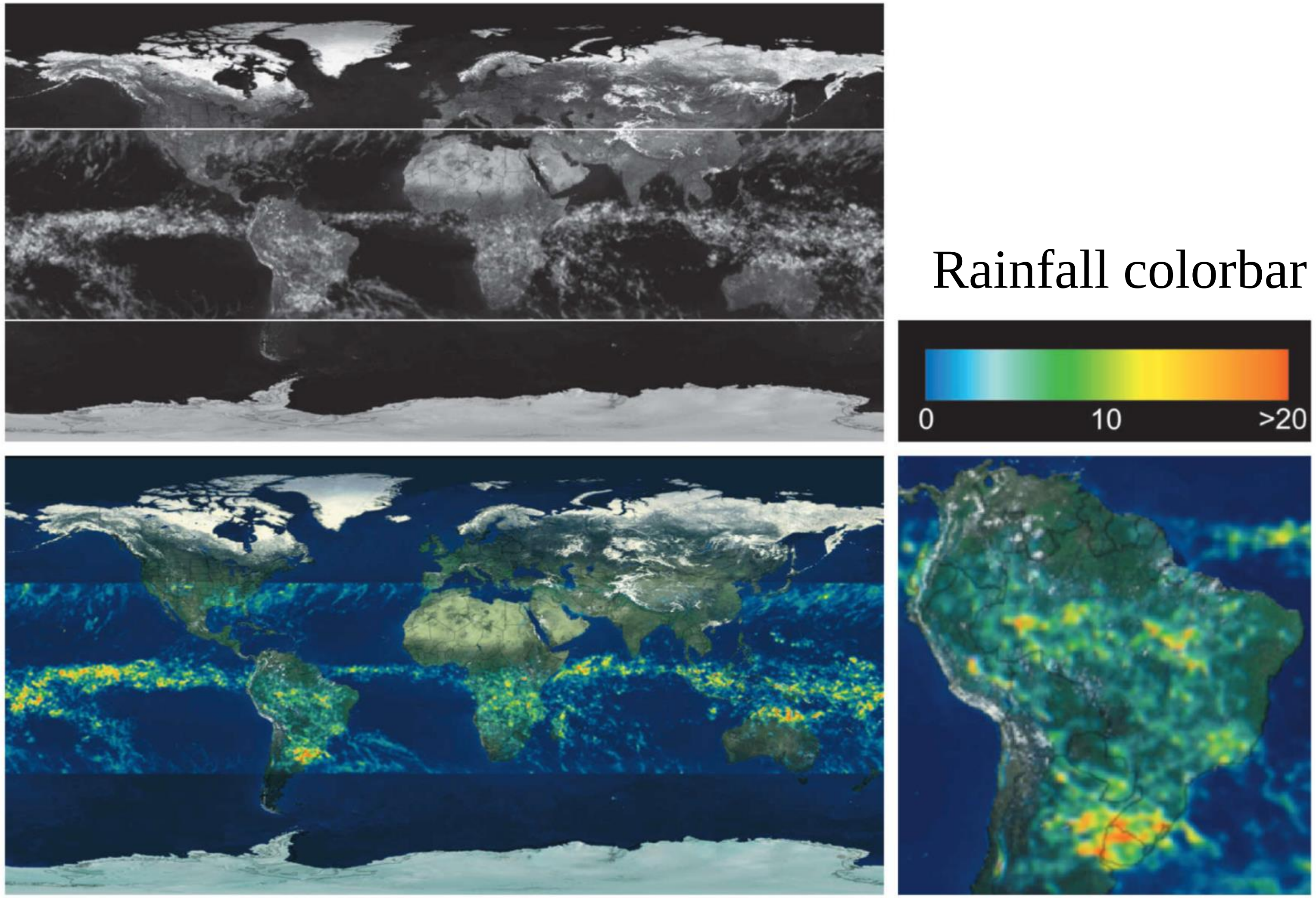
intensity slicing using eight colors



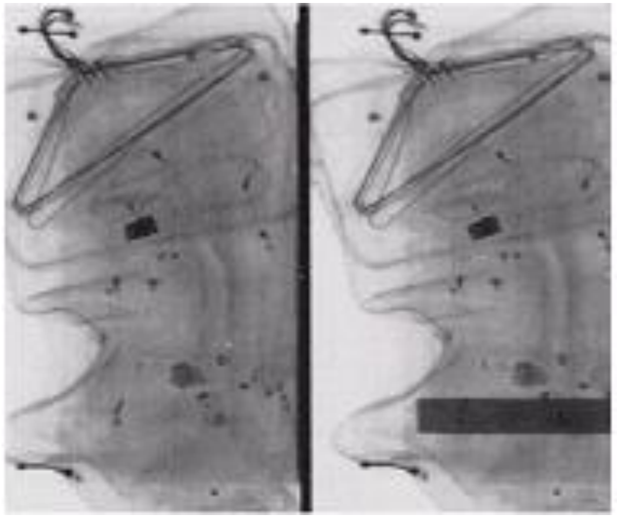
Pseudocolor

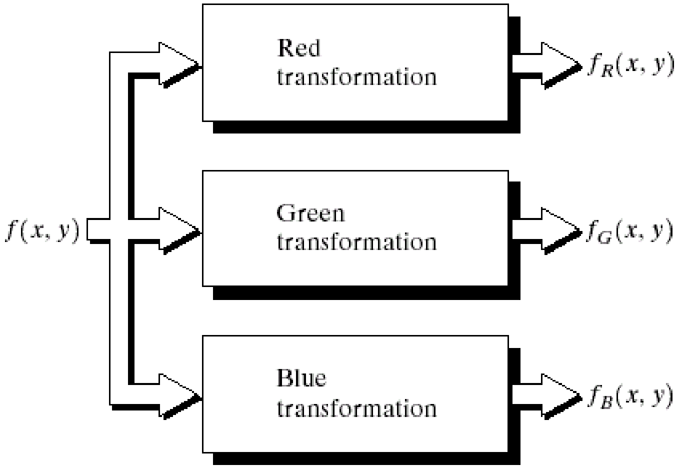
Colormap Name	Color Scale
parula	
turbo	
hsv	
hot	
cool	
spring	
summer	
autumn	
winter	
gray	
bone	
copper	
pink	
jet	
lines	
colorcube	
prism	
flag	
white	

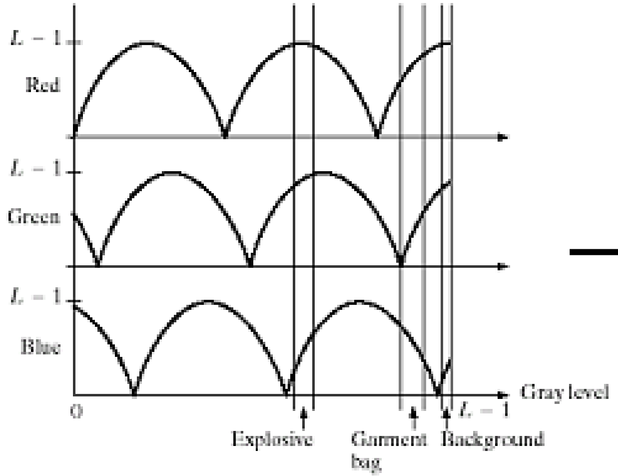
Pseudocolor

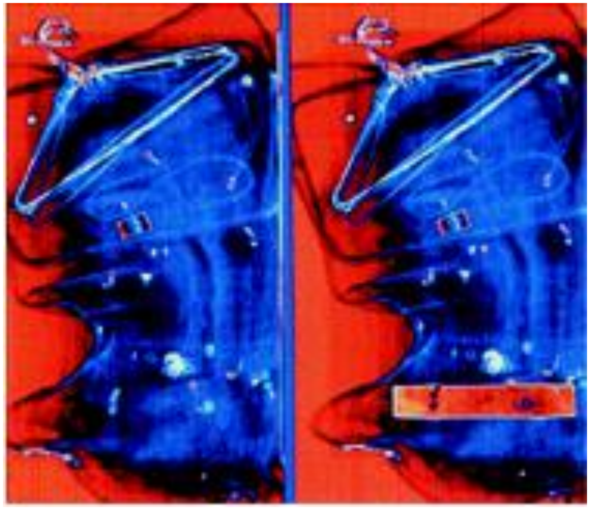


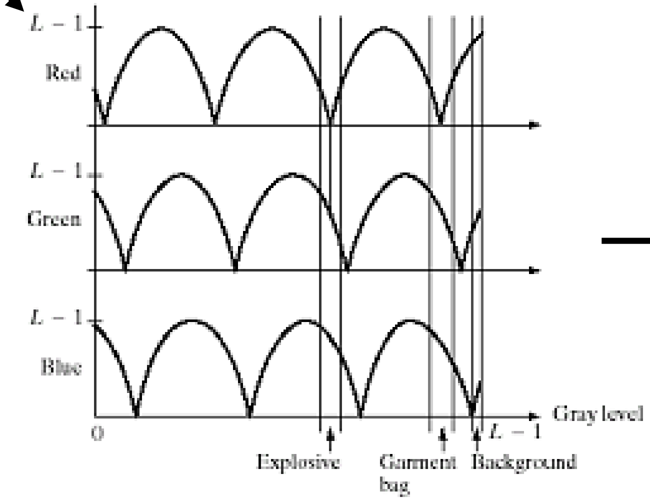
Pseudocolor

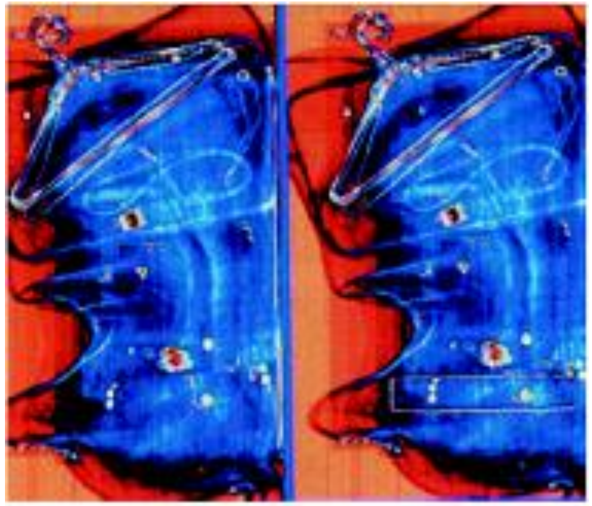








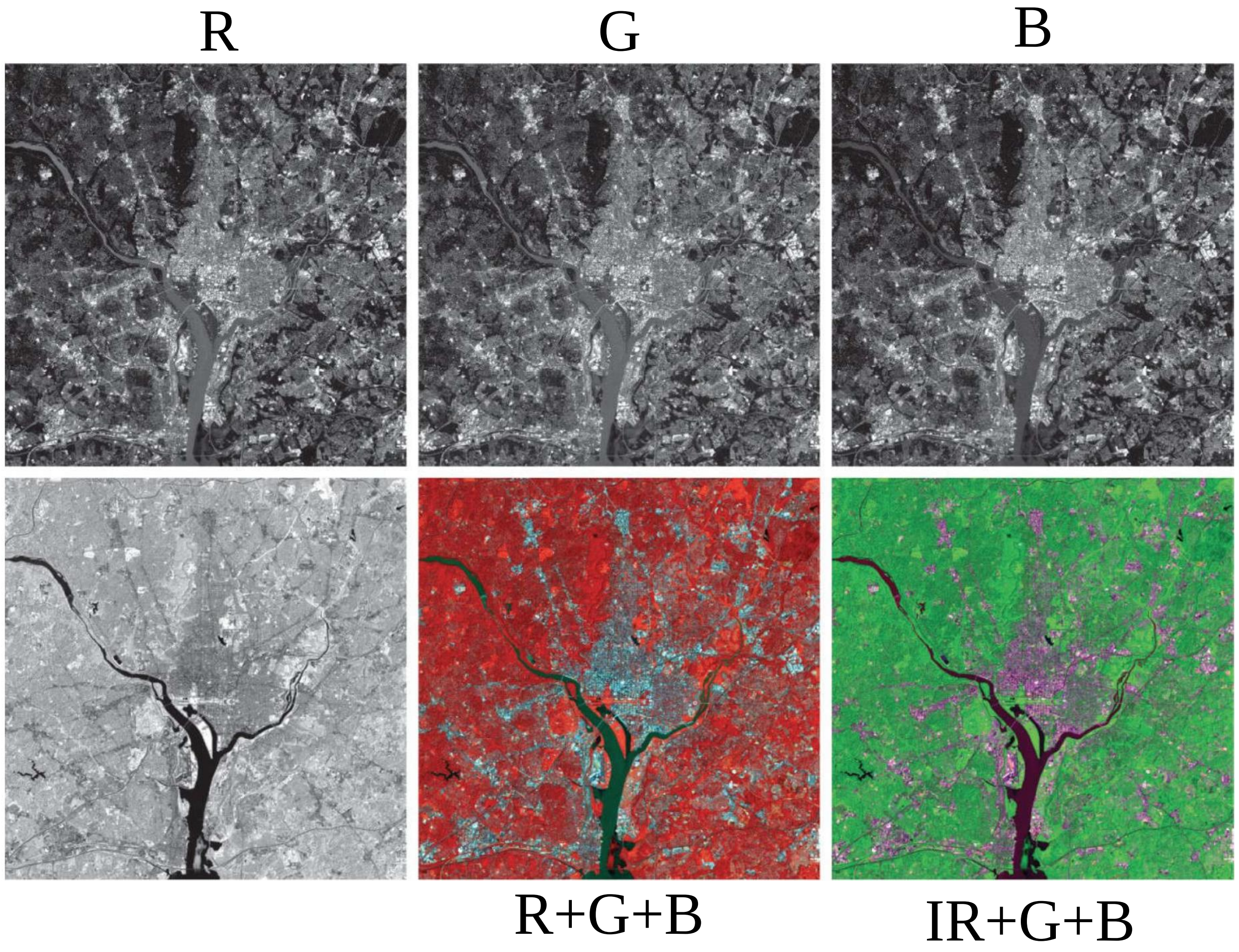




Pseudocolor

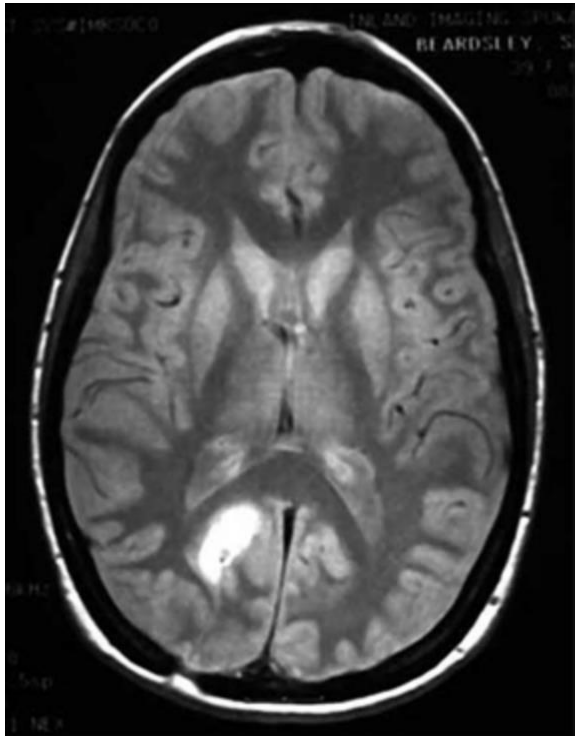
- Multi-modelity

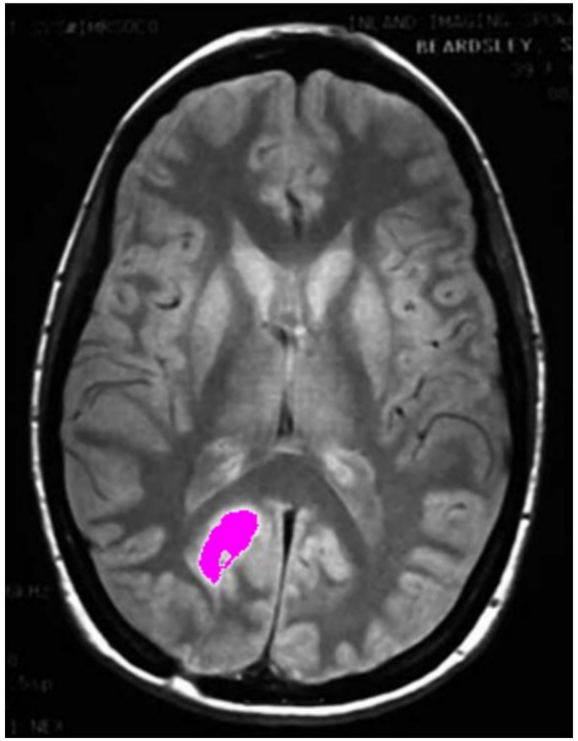
near infrared (IR)
biomass content



Pseudocolor

- False color
  - – Provide richer information than a gray-scale image
  - – Assigning colors to labels





Pseudocolor

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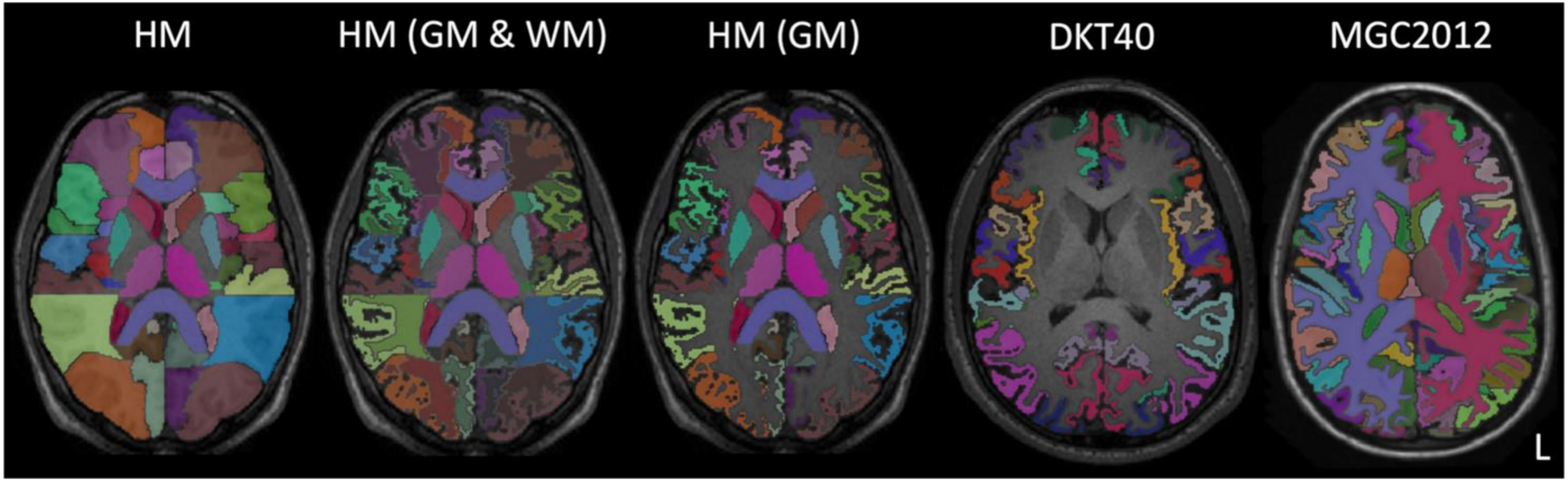
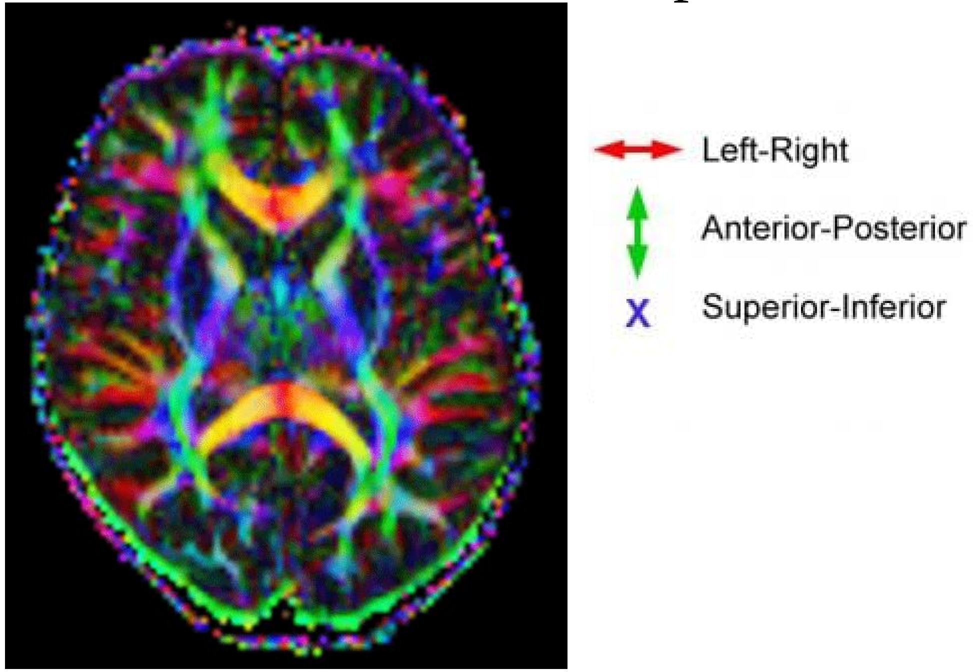


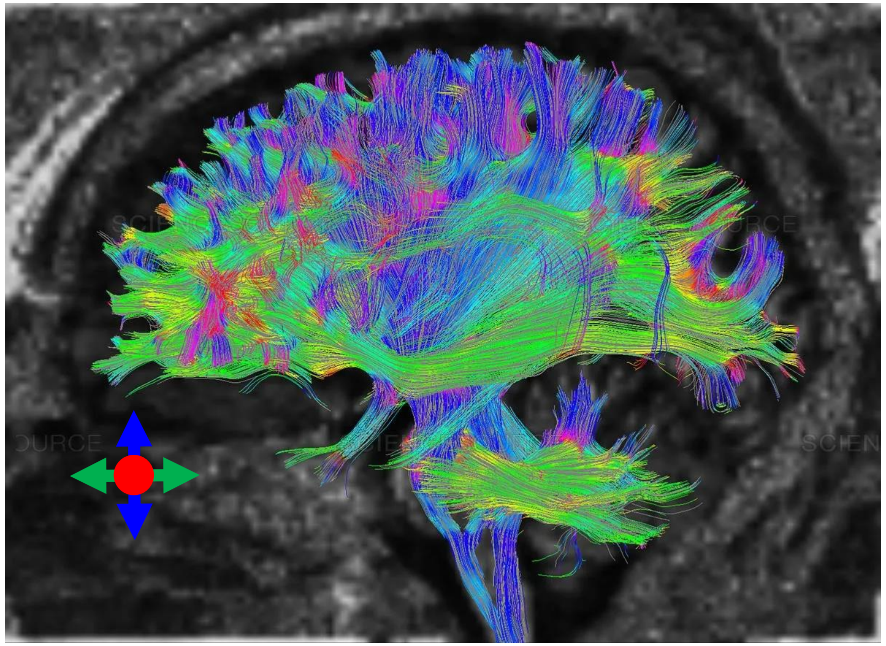
Pseudocolor

- False color
  - Provide richer information than a gray-scale image
  - Assigning colors to directions

Directional encoded colormap

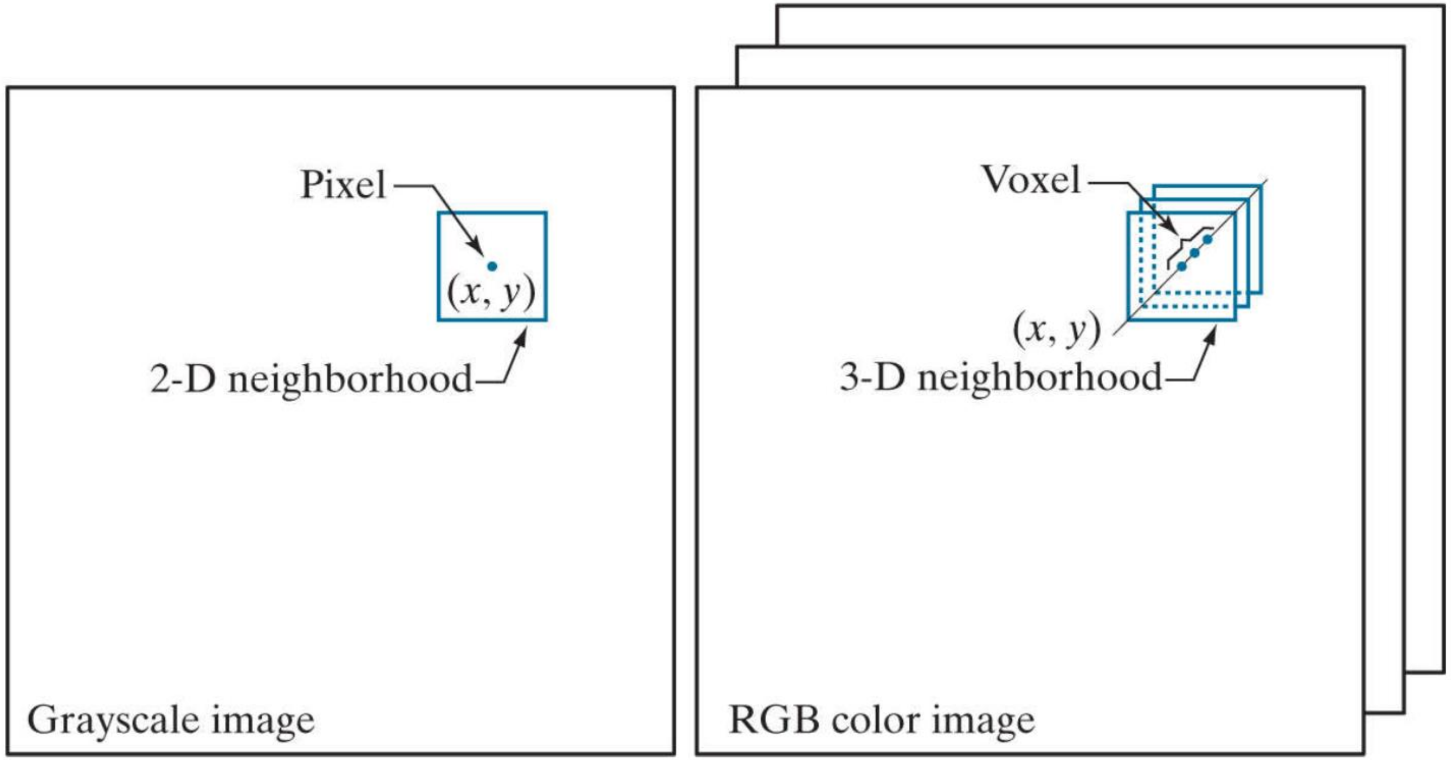


Tractography



Color Image Processing

- Similar to grey scale images



NxM

NxMx3

Outline

- Basic Image Concepts
  - Sampling and quantization
  - Interpolation

- Color Image Processing (Chapter 6)
  - Color model (RBG, CMY, HSI)
  - Pseudocolor (visibility, richer information)